Chapter 7

Optical Linewidth Measurements on Photomasks and Wafers†

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I. INTRODUCTION

Linewidths and other critical dimensions must be known and accurately controlled to enable integrated circuit (IC) performance to be predicted from design specifications, to facilitate transfer of accurate photomask dimensions between manufacturing and using organizations, and to monitor lithographic and patterning processes. Traditional methods for making linewidth measurements on photomasks and wafers have not been able to achieve the accuracy or precision required for VLSI geometries. However, the required tolerances of 10% or less on 1- to 2-μm line geometries can be met with optical linewidth-measurement techniques by using more careful adjustment, calibration, and control methods and by increasing the degree of automation.

In this chapter the origins of systematic errors in optical linewidth-measurement systems will be discussed, the advances in modeling the linewidth-measurement process will be outlined, including imaging in the optical microscope, a primary linewidth-measurement system in use at the National Bureau of Standards (NBS) will be described, and the use of primary measurements to calibrate less accurate systems conventionally used for linewidth measurements will be discussed. Although emphasis will be placed on measurements of patterns on antireflective (AR) chromium photomasks in transmitted light, measurements on other types of materials including "see-through" photomasks and wafers will also be discussed.

II. LIMITATIONS OF TRADITIONAL METHODS

Traditionally, the most widely used linewidth-measuring instruments have been based on optical imaging methods that employ a microscope in conjunction with some type of measuring attachment. These attachments may be grouped into one of the following categories: (1) filar, (2) image-shearing, and (3) image-scanning. In each category, the optical system has a video (TV) counterpart. For instance, the filar or micrometer eyepiece has an equivalent video micrometer in which the filar or cross hair is generated on the video monitor and positioned electronically. Image-shearing systems perform the shear optically, but the display may be either optical or video. Image-scanning systems employ either a photomultiplier and scanning slit or a video system. In this chapter, the photomultiplier system is referred to as an optical image-scanning system, whereas a semiautomatic TV-microscope system, which determines linewidth from analysis of the video image profile, is referred to as a video image-scanning system.

Other linewidth-measuring instruments are based on nonimaging optical techniques (which employ measurement in the Fourier spectrum plane or on a nonconventional image after some kind of optical processing, such as spatial filtering) or on electron imaging techniques.

The accuracy of optical imaging methods is dependent upon the characteristics of the optical image formed by the microscope and is affected by diffraction and aberrations in the optical system as well as the partial coherence of the illumination [1]. As shown in Fig. 1, even with very steep material edges, the optical image contains a gradual transition from light to dark at the line edge due to diffraction. The width of this transition region or image edge is comparable with the diffraction spot, or Airy disk diameter, of the imaging optics, approximately 1.0 μm for a 0.65 objective numerical aperture (N.A.) or 0.7 μm for a 0.90 N.A. To determine linewidth to an accuracy better than this, one must use an optical threshold that corresponds to the true line edge location. The importance of the choice of a correct threshold for linewidth measurement is illustrated in Fig. 2. An incorrect threshold produces an error in edge location. This error may cancel in a pitch measurement [2] if both lines have the same image profile, whereas the right and left edge errors add to produce a

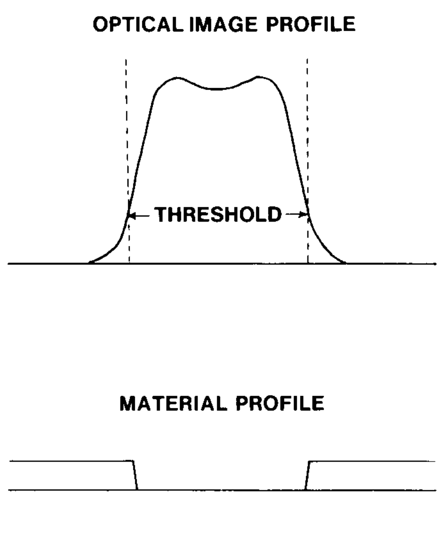


Fig. 1. Schematic representation of the relationship between the material edges and the optical image.

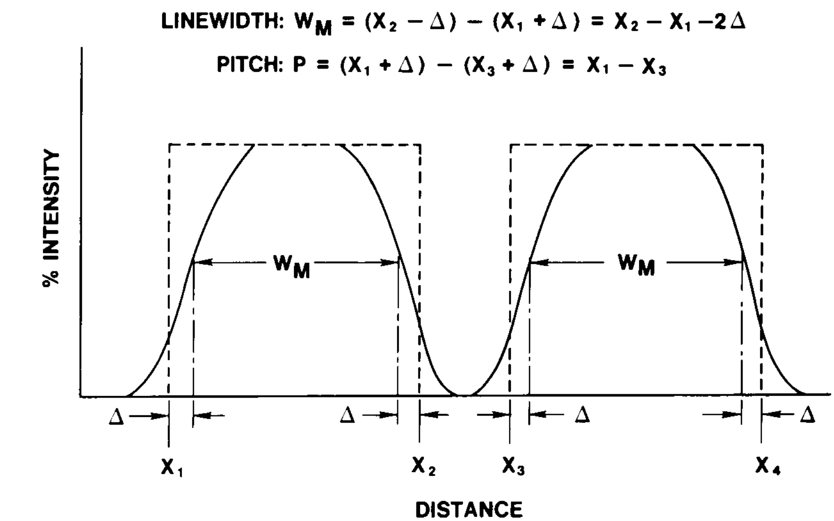


Fig. 2. The effect of the edge-detection error (Δ) caused by use of an incorrect threshold on pitch and linewidth measurements.

larger error in the case of a linewidth measurement. For this reason, line-scale calibration is inadequate to ensure accurate linewidth measurements.

The threshold used for edge detection varies considerably with the type of measuring instrument used [3–6]. As shown in Fig. 3, modeling of a

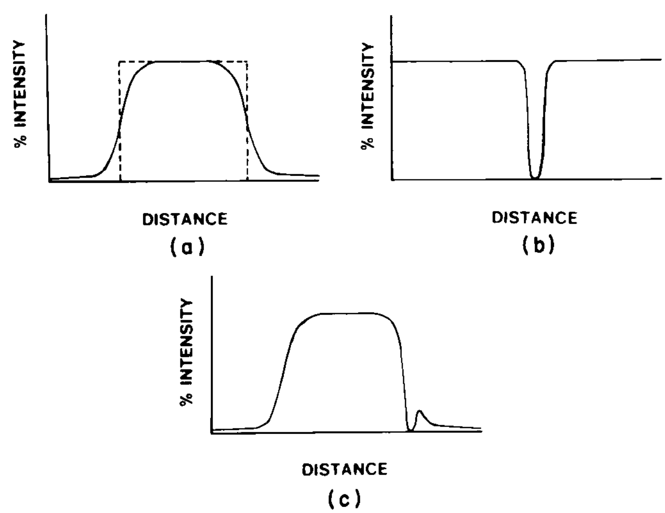


Fig. 3. Model of a filar eyepiece: The line-image profile (a) is multiplied by the transmittance profile of the cross hair (b) to produce the observed intensity distribution (c).

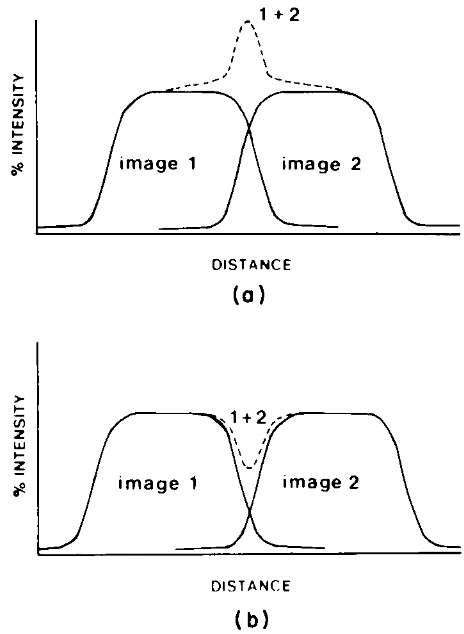


Fig. 4. Model of an image-shearing eyepiece: The addition of overlapping optical image intensities may produce either a bright line when the image profiles intersect above the 50% threshold (a) or a dark line when they intersect below the 50% threshold (b).

filar or micrometer eyepiece indicates the use of a very low threshold (less than 10%) corresponding to a just perceptible bright band to the dark side of the filar or cross hair. In contrast, an image-shearing eyepiece forms a second displaced image, which overlaps the first image as shown in Fig. 4. For a linewidth measurement, the amount of overlap is usually set to allow neither a bright line nor a dark band between the images; with incoherent illumination, this setting corresponds to a 50% threshold. A video image-scanning system uses a vidicon to generate a voltage proportional to the optical intensity. The linewidth is then determined by use of a voltage threshold for edge detection. Distortion of the line-image profile can result from the limited range (clipping) and slope (gamma) of the linear portion of the voltage response curve as shown in Fig. 5. Unless the video controls are properly adjusted, the voltage threshold corresponding to the line edge location may vary from day to day, resulting in a loss of both accuracy and precision.

The relationship between edge location and the parameter used to measure linewidth is more complex in nonimaging instruments. In coherent Fourier spectrum methods [7,8], a parameter, such as the distance between two minima, in the plane of the coherent spectrum is measured

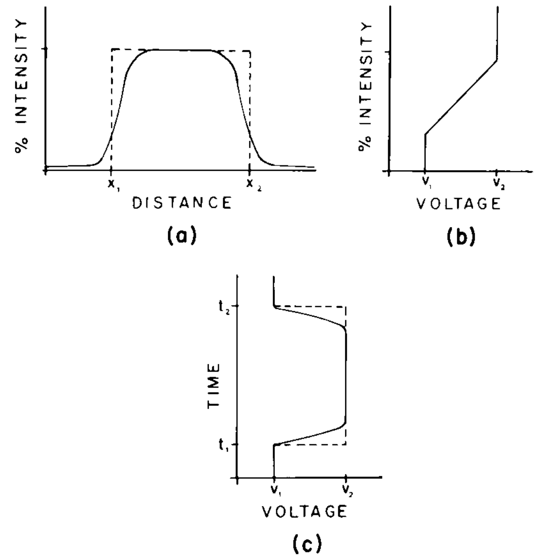


Fig. 5. Model of a video image-scanning system: The clipping levels (V_1 and V_2) and gamma (slope) of the voltage response curve (b) may distort the image profile (a) and change the relationship between the (electrical) threshold and edge location (c).

as shown in Fig. 6. Another method, from Swing [9], employs spatial filtering, which alters the image as shown in Fig. 7, in this case producing dark bands along the edges of the line being measured. In contrast to spatial filtering, dark-field microscopy [10,11] produces a bright band at the line edge, as shown schematically in Fig. 8. In this case, especially for lines narrower than 2 μm, the bright line may have an asymmetric profile or be slightly displaced from the edge. Such effects are dependent upon the numerical apertures of the optics employed [12].

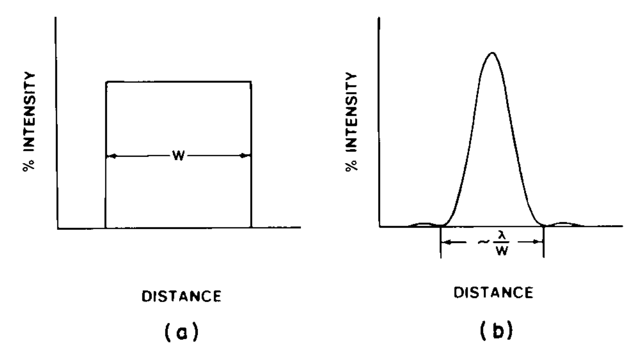


Fig. 6. Schematic of the relationship between the line object (a) and its coherent Fourier spectrum (b).

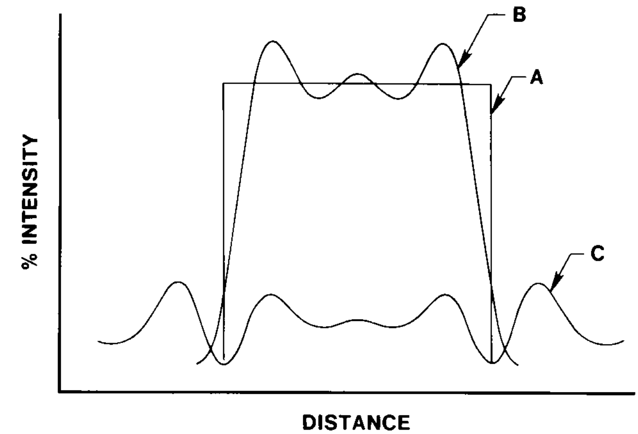


Fig. 7. Schematic of the relationship between the line object (curve A), the coherent image (no spatial filtering) (curve B), and the spatially filtered coherent line image (curve C).

Although this discussion is directed at optical measurement techniques, it should be noted that the accuracy of electron microscope imaging measurements also depends upon the quality and characteristics of the (electron) image [13] as well as the method of recording or analyzing that image, including the criterion used for edge determination [14].

Given the diversity of edge-detection criteria employed by different systems, it is not surprising that differences between measurements on different systems may be as large as 1 μm. These differences in linewidth measurements exist between different types of linewidth-measuring

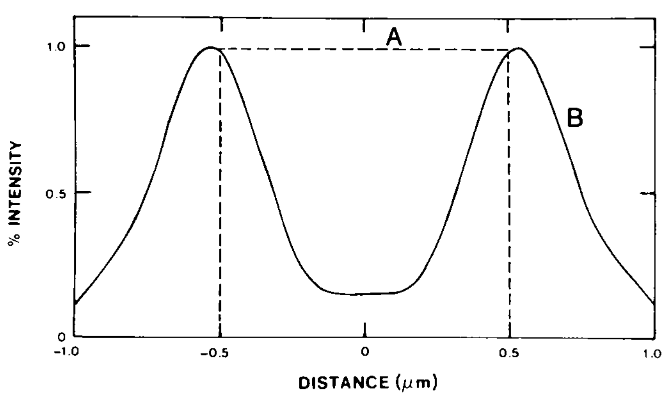


Fig. 8. Schematic of the relationship between the line object (dashed lines A) and the image produced in a dark-field microscope (solid curve B).

systems, between different systems of the same type, between different operators on the same system, and between measurements made by the same operator at different times of the day. Some of these differences may be attributed to manufacturing tolerances or human error, but a much larger contribution comes from the very real differences in the way that the various systems detect edges and determine linewidth.

These differences are illustrated in Table I, which shows a sampling of measurements made by different optical imaging systems on the same nominally 1- and 3-μm lines and spaces on an AR-chromium photomasklike artifact. Prior to use, each measuring system was adjusted to agree only with known center-to-center line spacings on a calibrated length scale. It is clear that differences in the magnitude observed become intolerable at VLSI dimensions. In order to determine the major sources and magnitudes of linewidth-measurement differences between instruments, as well as to develop more accurate measurement systems and calibration methods, a more detailed model of the optical imaging measurement process was developed [15–18]. The components of the model include an accurate description of the object to be measured, a partial coherence description of the microscope illumination [1], a description of the microscope imaging optics including diffraction and aberrations, and a description of the edge-detection criteria employed by the measuring eyepiece or photometric sensor.

TABLE I

Comparisons of Linewidth Measurements of the Same Nominal 1- and 3-μm Lines and Spaces on an Antireflective Chromium Photomask

	Linewidth (μm)			
Type of system	Spaces		Lines	
Photometric	0.98	3.13	0.87	2.87
Filar eyepiece	1.05	3.17	0.73	2.68
Filar eyepiece	1.44	3.71	0.66	2.78
TV Filar	0.85	3.05	1.14	3.00
Filar eyepiece	1.12	3.36	0.71	2.71
Shearing eyepiece	0.81	2.96	0.98	2.98
Filar eyepiece	1.55	3.68	0.94	2.88
Filar eyepiece	1.16	3.57	0.66	2.66
Shearing eyepiece	0.73	2.82	1.04	3.05
TV-image scanning	0.85	2.93	1.06	2.83
Shearing eyepiece	0.93	2.93	0.99	2.95
Photometric	1.00	3.10	0.75	2.80

III. MODELING OF THE OPTICAL IMAGE

A great deal of theoretical work has been done on the calculation of optical microscope imagery based on the scalar theory of partial coherence. In the late 1950s and early 1960s, several authors, including Richards and Wolf [19], Watrasiewicz [20,21], and Charman [22,23], published results that cast doubt on whether scalar coherence theory could be used to predict accurately the image profiles of objects viewed in transmission with high N.A. microscope objectives. Since this point is key to the ultimate accuracy that can be achieved for optical linewidth measurement, considerable effort has gone into examining this problem. Kinzly [24], without arriving at a definitive conclusion, shows that the image profiles measured by Charman could be attributed to a "degraded" edge object. In light of results obtained recently at NBS [15,18], Kinzly appears to have been, at least in part, correct. The characterization of the line or edge object has turned out to be one of the principal problems in producing agreement between the predictions of scalar theory and experimental measurements.

A. Image-Profile Calculations

In the simplest model of a line object, it is assumed that the edge consists of a discontinuous change in transmittance as well as a discontinuous change in the optical phase introduced on transmission through the object. Mathematically, in the scalar coherence theory formulation of optical imaging [25], the edge object is characterized in one dimension (1-D) by a complex amplitude-transmittance function

$$A(x) = t(x)e^{i\phi(x)},$$

where x is the position coordinate in the object plane, as shown in Fig. 9. The amplitude-transmittance function $t(x)$ is given by

$$t(x) = \begin{cases} 1, & x \leq 0, \\ (T_0)^{1/2}, & x > 0, \end{cases}$$

where T_0 is the ratio of the instrumentally measured transmittance in the nearly opaque region I_0 relative to that in the clear region I_M and the phase function $\phi(x)$ is given by

$$\phi(x) = \begin{cases} 0, & x \leq 0, \\ \phi_0, & x > 0. \end{cases} \quad (1)$$

The optical intensity I across the 1-D line image is given by

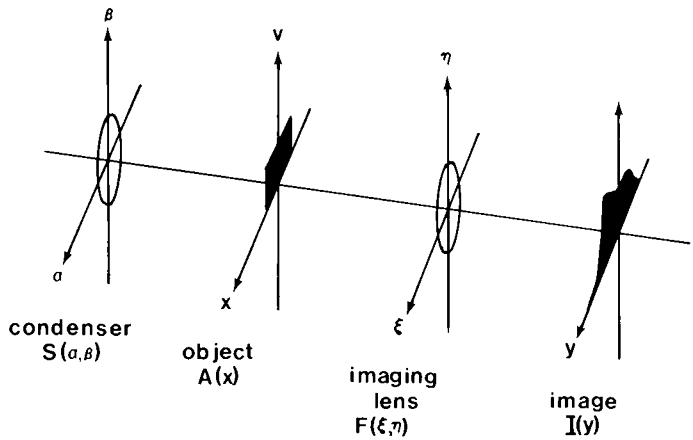


Fig. 9. Coordinates of the imaging system.

$$I(y) = \int_{-\infty}^{\infty} \tilde{I}(\xi) \exp[-2\pi i y \xi]\, d\xi, \tag{2}$$

where y is the position coordinate in the image plane and

$$\tilde{I}(\xi) = C \int_{-\infty}^{\infty} T(\xi + \xi',0;\xi',0)\tilde{A}(\xi + \xi')\tilde{A}^*(\xi')\, d\xi'. \tag{3}$$

The transmission cross coefficient [26, p. 530] for a 1-D object and a two-dimensional (2-D) optical system is

$$T(\xi_1,0;\xi_2,0) = \iint_{-\infty}^{\infty} S(\alpha,\beta)F(\xi_1 + \alpha,\beta)F^*(\xi_2 + \alpha,\beta)\, d\alpha\, d\beta, \tag{4}$$

where $S(\alpha,\beta)$ is the intensity distribution in the condenser aperture and F the 2-D pupil function of the imaging system. In Eqs. (2)–(4) and subsequent equations, the tilde (~) denotes the Fourier transform and the asterisk (*) the complex conjugate. The coordinates in the pupil plane of the imaging (objective) lens are ξ and η; see Fig. 9.

The spatial coherence at the object plane in 2-D is described by the mutual intensity $\Gamma(x_1,v_1;x_2,v_2)$, where (x_1,v_1) and (x_2,v_2) are pairs of points in the object plane.[1] For wide-field microscope illumination, where the edge object is uniformly illuminated over an area many times the Airy disk diameter of the condenser and the condenser aperture is incoherently

[1] In imaging systems, the usual time delay that appears in the mutual intensity is taken to be zero and has, therefore, been suppressed in the notation given here.

illuminated, the mutual intensity becomes a function of coordinate differences only [i.e., $\Gamma(x_1,v_1;x_2,v_2) = \Gamma(x_1 - x_2,v_1 - v_2)$]. The mutual intensity is related to $S(\alpha,\beta)$ through a Fourier transform:

$$\Gamma(x_1 - x_2,v_1 - v_2) = \iint_{-\infty}^{\infty} S(\alpha,\beta) \exp\{ik[\alpha(x_1 - x_2) + \beta(v_1 - v_2)]\}\, d\alpha\, d\beta, \tag{5}$$

where k is the mean wave number, assuming quasi-monochromatic illumination.

For the case where there is uniform illumination of both the condenser aperture and the edge object, the state of partial coherence may more simply be described by the coherence parameter R, the ratio of the numerical apertures of the condenser and objective lenses. Fully coherent illumination at the sample (characteristic of a well-collimated beam) corresponds to $R = 0$, whereas incoherent illumination (characteristic of an extended thermal source subtending a large cone angle) corresponds to $R = \infty$.

Note that for an optical scanning system with a fixed scanning aperture at the center of the image plane and a moving specimen or stage, the variable x becomes the displacement of the specimen or stage. Although it is possible to include the effect of the scanning slit, this correction is not necessary when the effective slit width is $\frac{1}{8}$ or less than the diameter of the Airy disk of the imaging objective lens [24].

For fully coherent illumination ($R = 0$), the foregoing analysis can be simplified considerably. In this case,

$$\begin{aligned} S(\alpha,\beta) &= \delta(\alpha)\delta(\beta), \\ \Gamma(x_1 - x_2,v_1 - v_2) &= 1, \\ T(\xi_1,0;\xi_2,0) &= F(\xi_1,0)F^*(\xi_2,0), \end{aligned}$$

and

$$\tilde{I}(\xi) = C \int_{-\infty}^{\infty} F(\xi + \xi',0)\tilde{A}(\xi + \xi')F^*(\xi',0)\tilde{A}^*(\xi')\, d\xi',$$

so that the optical intensity across the line image simplifes to

$$I(y) = C \left| \int_{-\infty}^{\infty} A(x)K(y - x)\, dx \right|^2, \tag{6}$$

where K is the complex amplitude-impulse response of the imaging system related to the pupil function through the Fourier transform

$$K(x) = \int_{-\infty}^{\infty} F(\xi,0) \exp[-2\pi i x\xi]\, d\xi. \tag{7}$$

Substituting for $A(x)$ in Eq. (6), one obtains the optical intensity at the image edge ($y = 0$):

$$I(0) = C \left| \int_{-\infty}^{0} K(x)\, dx + (T_0)^{1/2} \exp(i\phi_0) \int_{0}^{\infty} K(x)\, dx \right|^2 . \tag{8}$$

With the assumption only that the amplitude-impulse response has the appropriate symmetry,

$$\int_{-\infty}^{0} K(x)\, dx = \int_{0}^{\infty} K(x)\, dx = \tfrac{1}{2} \int_{-\infty}^{\infty} K(x)\, dx, \tag{9}$$

the optical threshold at the line edge is given (in the coherent case) by

$$\begin{aligned} T_c &= I(0)/I(y \to -\infty) \\ &= |0.5[1 + (T_0)^{1/2} \exp(i\phi_0)]|^2 \\ &= 0.25(1 + T_0 + 2(T_0)^{1/2} \cos \phi_0). \end{aligned} \tag{10}$$

The threshold is thus the 25% point corrected for T_0 and ϕ_0. It should be observed that the commonly used optical linewidth-measuring instruments do not employ this value of threshold nor do they allow for variations in threshold with variations in phase difference or spatial coherence. The threshold of Eq. (10) is also appropriate for use in partially coherent illumination when $R \leq 0.67$ for high-contrast masks such as AR-chromium and when $R \leq 0.25$ for low-contrast masks or wafers. This defines the conditions for effective coherence and can be demonstrated with calculations of the complete image profile [15,18]. These calculations can conveniently be carried out numerically by Kintner's method [27]. Prior to the introduction of this fast method for computing partially coherent images based on the transmission cross-coefficients [26, p. 530], it had been prohibitively time-consuming to include the optical phase in these calculations.

The method of calculation assumes a periodic object of period P, shown in Fig. 10; the period may be made arbitrarily large to permit analysis of isolated objects. The line object is then represented by a Fourier series

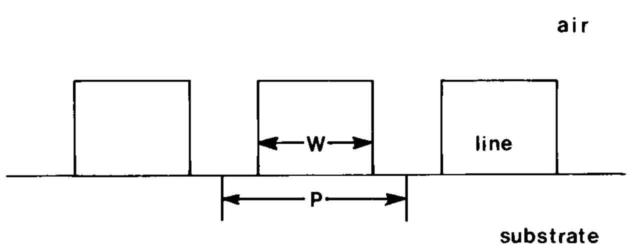


Fig. 10. Schematic of a periodic line object.

$$A(x) = \sum_m a_m \exp[2\pi i \omega_0 m x], \tag{11}$$

where $\omega_0 = 1/P$. If the line is chosen to be symmetrically located with respect to the origin $x = 0$,

$$A(x) = \sum_m a_m \cos(2\pi \omega_0 m x), \tag{12}$$

where $a_{-m} = a_m$. The optical intensity in the image becomes a discrete summation

$$I(y) = C_0 + 2 \sum_{m=1}^{\infty} C_m \cos(2\pi m \omega_0 y), \tag{13}$$

where

$$\begin{aligned} C_m = {} & a_m a_0^* T(m\omega_0,0;0,0) \\ & + \sum_{m'=1}^{\infty} \{a_{m+m'} a_{m'}^* \cdot T[(m + m')\omega_0,0;m'\omega_0,0] \\ & \qquad + a_{m-m'} a_{-m'} \cdot T[(m - m')\omega_0,0;-m'\omega_0,0]\}. \end{aligned}$$

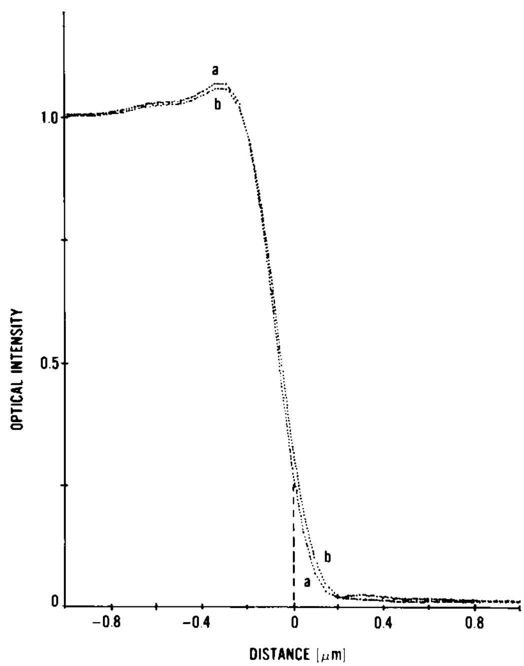


Fig. 11. Comparison of theoretical edge image profiles with (curve a) and without (curve b) $\pi/2$ phase: $T_0 = 0.01$, 0.9 objective N.A., 0.6 condenser N.A., and 530-nm wavelength.

The effect of including the optical phase step in the model of the imaging system is illustrated in the computed image profiles of Figs. 11–13. In each case, the effect of a $\pi/2$ phase difference is compared with that for no phase for the same optical system parameters. For example, with a background relative transmittance of 1% (Fig. 11), typical of patterns in AR-chromium photomasks, the difference in the measured linewidth with the assumption of no phase in the threshold formula and with the assumption of $\pi/2$ phase is 0.02 μm. However, for 40% relative reflectance, typical of lines etched in silicon dioxide on silicon, the error is much larger. The edge-detection threshold given by Eq. (10) for this case (Fig. 13) is 67% without phase and 41% with $\pi/2$ phase. If the 67% threshold is used, the presence of $\pi/2$ phase would result in an error of more

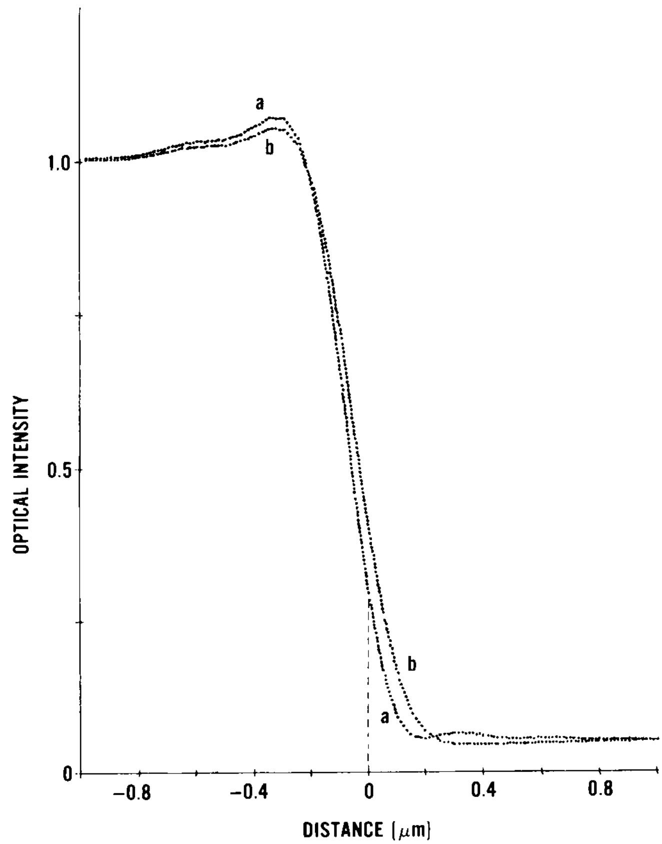


Fig. 12. Comparison of theoretical edge image profiles with (curve a) and without (curve b) $\pi/2$ phase: T_0 = 0.05, 0.9 objective N.A., 0.6 condenser N.A., and 530-nm wavelength.

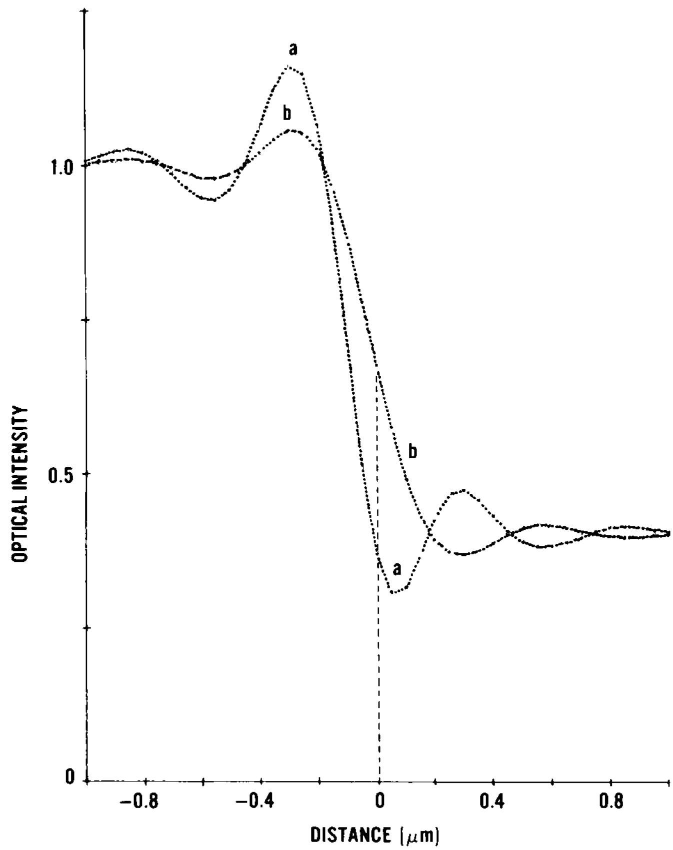


Fig. 13. Comparison of theoretical edge image profiles with (curve a) and without (curve b) $\pi/2$ phase: $T_0 = 0.4$, 0.95 objective N.A., 0.25 condenser N.A., and 530-nm wavelength.

than 0.1 μm in the linewidth measurement for the image profile shown. The use of 0 and $\pi/2$ phase steps is merely illustrative; the true phase step depends on the materials of the object and must be determined for each case as will be discussed in Section VII.

B. Focus Criteria

The choice of focus criterion is also very critical for accurate linewidth measurement. For high-contrast AR-chromium photomasks, several different focus criteria are possible. Figure 14 shows the calculated image profile for three focus positions. One possible focus criterion is maximum overshoot at the line edge corresponding to a bright band along the edge.

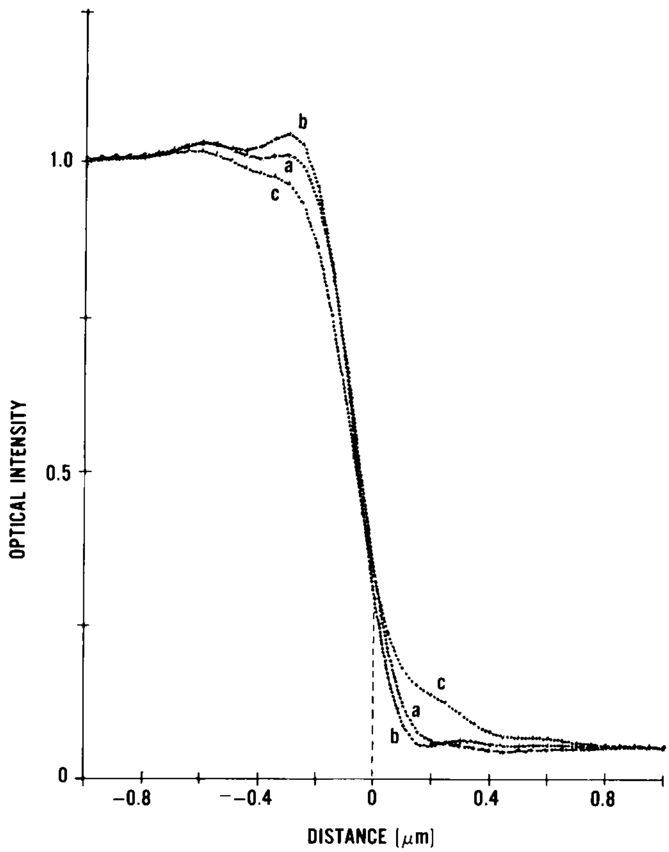


Fig. 14. Change in theoretical edge image profiles with focus parameter a_2 [see Eq. (16)]: curve a $-\lambda/4$; curve b, 0; and curve c, $+\lambda/4$, $T_0 = 0.05$, $\phi_0 = \pi/2$, 0.95 objective N.A., 0.6 condenser N.A., and 530-nm wavelength.

For visual measurements, an operator must be very well trained to see this band since it is barely observable visually and disappears with very slight defocusing. In the presence of spherical aberration, this bright band may be enhanced, and maximum overshoot can occur when the image is distinctly out of focus. Another focus criterion is the presence of the very faint interference band on the dark side of the line edge. This band disappears very quickly with defocus and may not be observable at all if there is too much flare light in the system. For linewidths or spaces greater than 1 μm, use of the criterion of steepest slope is the best method of determining best focus; this criterion is easily adapted to automatic focusing systems.

For other line objects such as see-through photomasks or patterns in silicon dioxide on a silicon wafer, the problem of accurate focus becomes much more complex. The line image seen in the microscope is asymmetric with focus position. As shown in Fig. 15, the dark band at the line edge is enhanced to one side of the best focus, whereas the bright band is enhanced with displacement in the opposite direction. Under these lower-contrast conditions (a larger influence of the relative phase difference on the image structure), maximum intensity at the line edge does not correspond to best focus. In this case, steepest slope also appears to yield the best objective criterion for autofocusing systems.

The asymmetry with focus, which may or may not occur depending upon the relative phase contribution to the image, deserves some discussion. With the assumption of equal lines and spaces for simplicity, the Fourier series representation of a multiple-line object given by Eq. (11) becomes

$$A(x) = \frac{1}{2}(1 + (T_0)^{1/2}e^{i\phi_2}) + \sum_{m\neq 0} \frac{\sin(\pi m/2)}{\pi m}(1 - (T_0)^{1/2}e^{i\phi_2}) \exp(2\pi i\omega_0 mx), \qquad (14)$$

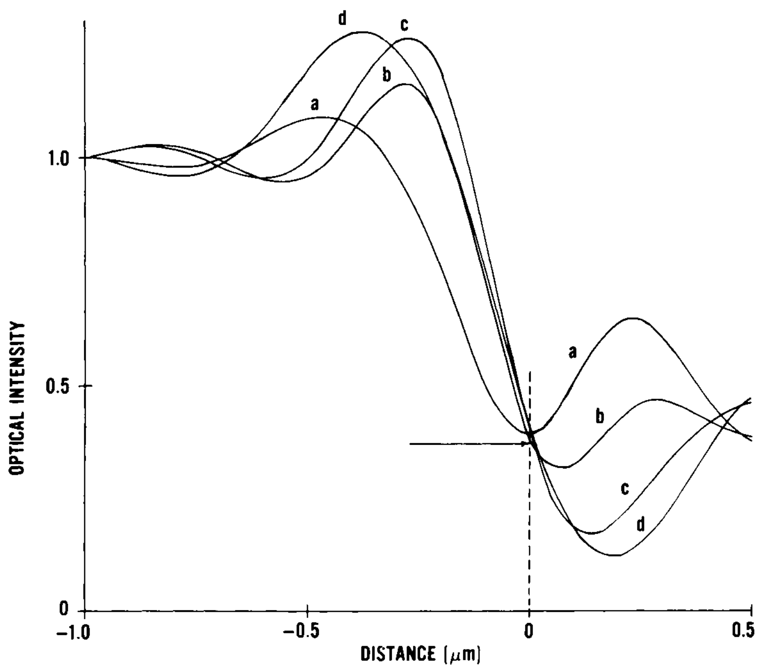


Fig. 15. Change in theoretical edge image profiles with focus parameter a_2 [see Eq. (16)]: curve a, $-\lambda/4$; curve b, 0; curve c, $+\lambda/4$; curve d, $+\lambda/2$; $T_0 = 0.4$, $\phi_0 = \pi/2$, 0.95 objective N.A., 0.25 condenser N.A., and 530-nm wavelength.

where the Fourier coefficients a_m of Eq. (11) have been computed for the periodic line object shown in Fig. 10. The coherent amplitude Fourier spectrum in the pupil plane of the imaging lens is then given by

$$\tilde{A}(\xi) = \frac{1}{2}\,(1 + (T_0)^{1/2}e^{i\phi_2})\,\delta(\xi) + \sum_{m\neq 0} \frac{\sin(\pi m/2)}{\pi m}\,(1 - (T_0)^{1/2}e^{1\phi_2})\,\delta(\xi - m\omega_0). \tag{15}$$

A diffraction-limited lens with only defocus present has a pupil function of the form

$$F(\xi) = \mathrm{Rec}(\xi|M\omega_0)\,\exp(ika_2\xi^2), \tag{16}$$

where $\mathrm{Rec}(\xi|M\omega_0)$ is the rectangular function in the coordinate ξ of width $2M\omega_0$ and a_2 the amount of defocus in wavelengths. Multiplying $F(\xi)$ by $\tilde{A}(\xi)$ gives the amplitude spectrum of the image

$$\tilde{A}_I(\xi) = \frac{1}{2}\,(1 + (T_0)^{1/2}e^{i\phi_2})\,\delta(\xi) + \sum_{m\neq 0} \frac{\sin(\pi m/2)}{\pi m}\,\delta(\xi - m\omega_0) \times \{\exp(ika_2\xi^2) - (T_0)^{1/2}\exp[i(\phi_2 + ka_2\xi^2)]\}. \tag{17}$$

For a high-contrast ($T_0 = 0$) line object, the second term in braces disappears, producing a coherent symmetric-with-focus image. If $\phi = 0$, the term in braces becomes $(1 - (T_0)^{1/2})\exp(ika_2\xi^2)$, which produces the same amplitude image structure as the high-contrast case multiplied by the constant $1 - (T_0)^{1/2}$. If $\phi = \pi$, we can make use of the relationship.

$$e^{i(\pi+x)} = -e^{ix}$$

to show that the term in braces in Eq. (17) becomes $(1 + (T_0)^{1/2})\exp(ika_2\xi^2)$. This result also produces the same amplitude image structure as the high-contrast case (symmetric with focus) multiplied by the constant $1 + (T_0)^{1/2}$. For all other cases, the sign of a_2 becomes important when added to ϕ and produces the asymmetry with respect to focus that makes accurate focus difficult for both visual and automated focusing. With linewidth-measurement systems that do not detect edge position accurately and require linewidth calibration, emphasis should be placed on repeatability of the focus criterion used. Steepest slope appears to be the most repeatable method for automated systems.

IV. PRIMARY LINEWIDTH-MEASUREMENT SYSTEM

An optical image-scanning microscope system was designed and assembled at the National Bureau of Standards (NBS) to enable linewidth

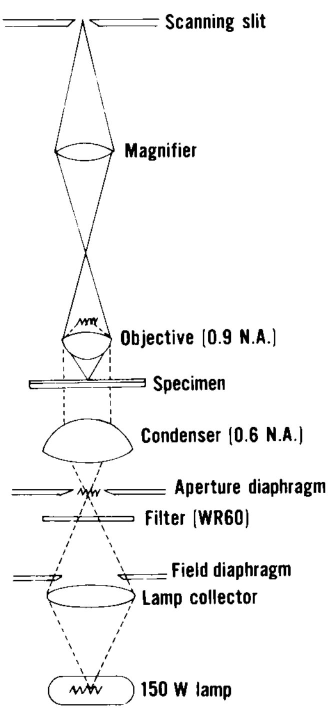


Fig. 16. Ray path for transmitted-light scanning microscope system (see text for discussion).

measurements to be made under conditions consistent with the assumptions and restrictions employed in developing the theory. The ray path for this system, which is built around a Zeiss Universal microscope,[2] is shown in Fig. 16. The light source is a 150-W tungsten–halogen lamp filtered for green light with a Wratten 60 (WR 60) filter[2] and a hot mirror[3] producing an effective spectral bandwidth of approximately 60 nm peaked at 530 nm, the principal wavelength at which most optical microscopes are diffraction limited. The condenser lens has a numerical aperture of 0.60, and the objective lens is a 63×, 0.90-N.A. Zeiss Planachromat designed for use without a cover glass ($R = 0.67$). This combination provides

[2] Throughout this chapter certain commercially available materials or instruments are identified for the purpose of providing a complete description of the work performed. The experiments reported do not constitute an evaluation of the performance characteristics of the products so identified. In no case does such identification imply recommendation or endorsement by the National Bureau of Standards nor does it imply that the items identified are necessarily the best available for the purpose.

[3] The hot mirror is required for photometric systems to provide a long-wavelength cutoff; the limited sensitivity of the eye performs the same function for visual systems.

a diffraction-limited system with effectively coherent illumination at the object plane; the high-N.A. objective is required to achieve the best possible resolution. The overall system magnification is 158×.

The scanning aperture is a 20 × 200-μm slit; the effective slit width at the object plane is 0.13 μm so that the slit has negligible effect on the imagery. Contrary to conventional use, the scanning slit is deliberately chosen smaller than the diameter of the impulse response of the imaging optics in order to preserve detail in the image that relates to the structure of the Airy disk and yields information about the edge location. The scanning slit is fixed; scanning is achieved by moving the object stage instead of image scanning so that the optical system is always used axially and is therefore devoid of off-axis aberrations such as coma, distortion, and curvature of field. The line object is moved in a direction perpendicular to the scanning slit by means of a piezo-driven flexure-pivot stage [28] mounted on the microscope stage; this stage has subnanometer resolution, smooth motion at the 3-nm level, and roll and pitch of less than 0.1 arc sec. Thus it is well suited to scanning small objects in the 0.5–10-μm range. The amount of stage displacement in one dimension is determined by a laser interferometer.

Fine-focus adjustment is provided by a piezoelectric ring stack mounted between two plates [18]. A hole in the center of the ring stack and corresponding holes in the plates allow the condenser lens to be brought up through the ring stack into focus at the object plane for transmitted light measurements. The lower plate is supported by the flexure-pivot stage, and the photomask to be measured rests on the upper plate. Focusing accuracy to better than 0.1 μm is achievable with this device. Because of hysteresis in the piezoelectric elements, one must go through focus in one direction only or return to zero voltage before incrementing or decrementing the voltage. This extremely smooth, fine motion allows an enormous improvement over conventional mechanical fine-focus adjustments.

The detector in the optical image-scanning microscope system is an ac photometer with an accuracy of 1% over a density range of 2.0 D when operating at 350 Hz with a maximum total collected flux of approximately 1 nW. The photomultiplier is the bi-alkali type (EMI 9635B[4]) chosen for maximum sensitivity with least noise. To reduce the effects of vibration, the system is mounted on a 1500-lb vibration-isolation table. With this system, a linewidth-measurement precision or repeatability (three times the sample standard deviation for samples of 30 or more measurements) of ± 0.04 μm can be achieved. It should be noted that a linewidth-

[4] See footnote 2 on p. 319.

measurement system of this type, which uses effectively coherent illumination, is actually a microdensitometer operating in a controlled nonlinear mode [29] that has been optimized for edge detection. It produces the maximum slope at the line edge and therefore the greatest sensitivity to small changes in linewidth.

V. PRIMARY LINEWIDTH MEASUREMENTS ON PHOTOMASKS

A typical comparison of calculated and measured edge profiles is shown in Fig. 17. The edge examined is in a 150-nm-thick AR-chromium layer on a glass plate. For calculation, 1% transmittance of the chromium, a $\pi/2$ phase difference, and an infinitely steep edge were assumed. The fit is

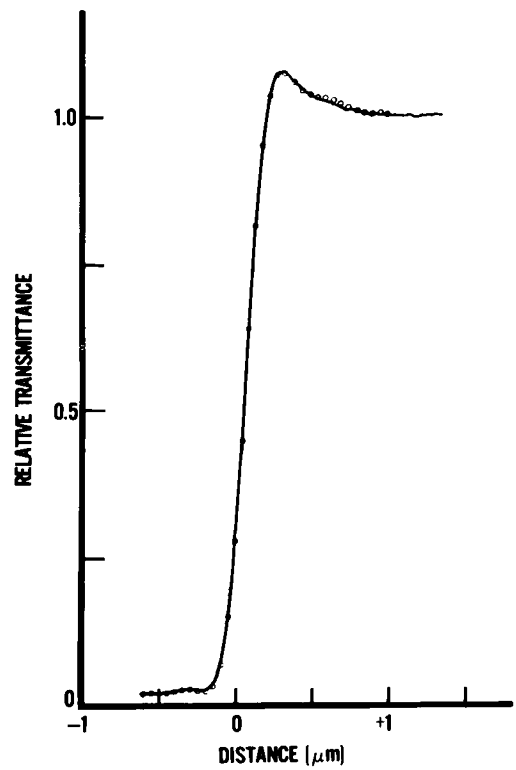


Fig. 17. Comparison of experimental (—) and theoretical (O) edge profiles for AR-chromium; $T_0 = 0.01$, $\phi_0 = \pi/2$, 0.9 objective N.A., 0.6 condenser N.A., and 530-nm wavelength.

made simply by matching the two curves at the threshold T_c. There is virtually no discernible difference between the two curves. Although this comparison shows the capability of the optical system, it is nearly impossible to reproduce all of the detail in the toe and shoulder of the image during routine measurements. Small imperfections in the edges coupled with small focus errors produce differences, typically less than 1%, in these regions.

The accuracy of the length scale of the optical image-scanning microscope is established by means of a line-spacing calibration [30], which is traceable to the national length standard with a precision of ± 10 nm. The accuracy of the edge-detection criterion is established through the high degree of agreement between the measured and theoretical images. This system thus serves as a primary calibration system for photomasklike reference linewidth standards [31].

It is also possible to establish the accuracy of the edge-detection criterion by means of scanning electron microscope (SEM) measurements [32]. While edge-detection errors occur in SEM measurements as they do in optical measurements, their magnitude for ideal vertical edges is typically at least a factor of five smaller. Some preliminary comparisons between optical and SEM measurements have been made [14], and with the exception of one anomaly, the agreement between the SEM and optical measurements was satisfactory. However, additional work on modeling of the SEM measurement system, prediction of image profiles, and development of edge-detection algorithms paralleling the optical work is necessary to assure the accuracy of the SEM measurements to the desired level. Such analysis is currently being developed by a number of researchers [33–37].

Despite the good agreement achieved between the calculated and measured profiles, it must be observed that the assumption of a vertical edge (infinite slope) is not realistic. From SEM photomicrographs of cross sections of AR-chromium edges and profilometer measurements, it is estimated that the AR-chromium film is approximately 150 nm thick with an edge slope that may deviate from the vertical by as much as 20°, resulting in an edge width Δ of up to approximately 0.05 μm, as illustrated schematically in Fig. 18. Despite the fact that the shape of the optical image does not appear to be affected significantly by the finite edge slope, there is an uncertainty of amount Δ regarding the position on the object, corresponding to the threshold value of transmittance used for edge detection. Because of this, ± 0.05 μm has been identified as the basic limit of accuracy of the measurement, although the precision characterized by three times the sample standard deviation for an average of nine or more repeated measurements is about ± 0.01 μm.

This limit of accuracy applies uniformly to lines with comparable edge

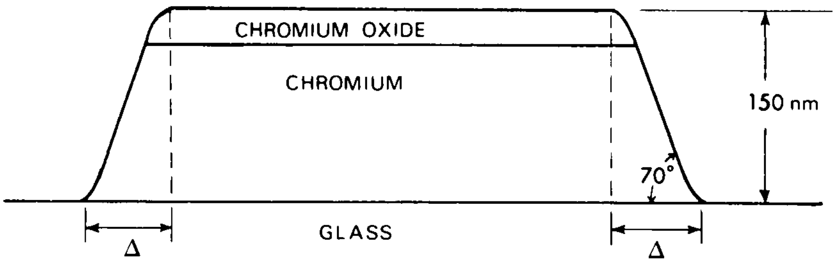


Fig. 18. Model of an AR-chromium photomask line object. The width (Δ) of the sloped portion of the material edge determines the lower limit on the uncertainty (±Δ) of the linewidth measurement.

slopes down to dimensions of 0.5 μm. The difference between the true linewidth and that resulting from the use of the edge-detection threshold given by Eq. (10) is less than 0.01 μm for dimensions greater than 0.5 μm. For smaller dimensions, the errors increase, as shown in Fig. 19, with a smallest measurable dimension of 0.4 μm. Extension to still smaller dimensions would require the use of either shorter wavelength illumination (<530 nm) in an optical microscope or an SEM.

VI. LINEWIDTH MEASUREMENTS WITH CONVENTIONAL OPTICAL SYSTEMS ON PHOTOMASKS

As noted in the introduction, most optical linewidth-measurement systems in use today are based on microscopy; the microscope is the

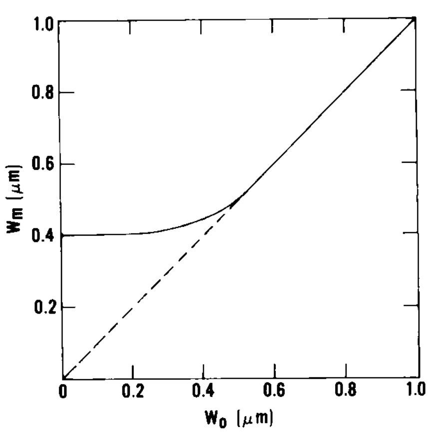


Fig. 19. Theoretical limit of resolution of the NBS measurement system. The linewidth W_m measured using the threshold T_c for edge detection versus true linewidth W_0 ($T_0 = 0.01$, $\phi_0 = \pi/2$, 0.9 objective N.A., 0.6 condenser N.A. ($R = 0.67$), and 530-nm wavelength). The dashed line represents the ideal system response.

basic building block to which any one of a variety of measurement attachments may be added. The accuracy and precision of these measurement systems are dependent upon the quality and reproducibility of the optical microscope image in addition to the edge-detection criteria discussed previously. Detailed procedures have been developed for preparation of microscopes for dimensional measurements on AR-chromium masks in such a way as to reduce the effects of variation of the image structure on the measurement.[5] The procedures provide for: a green filter; wide-field, bright-field, Kohler illumination, which provides uniform illumination of the object, reduces stray light, and results in bright images with good contrast; effectively coherent illumination, which provides an image with steep edge gradients; and proper illumination levels in both visual and optical scanning instruments. Other procedures[5] have been developed for adjustment and calibration of the linewidth-measuring attachments discussed here. These procedures were evaluated and modified as a consequence of two field studies conducted by the NBS with the cooperation of numerous manufacturers of photomasks, ICs, and linewidth-measurement equipment [38,39].

Examples of measurement data from different types of commercially available linewidth-measurement systems are discussed in the following sections. Measurements were made on AR-chromium photoplates patterned by conventional photolithographic techniques. The plates contained eight patterns, each of which contained lines and spaces in the range between 0.5 and 12 μm [40] together with pairs of lines for line-spacing calibration and pairs of lines and spaces for instrument adjustment of line-to-space ratios where necessary. The pattern is similar to the basic pattern of NBS Standard Reference Material SRM 474 [31] shown in Fig. 20. One of the patterns on each photoplate used was measured by an optical image-scanning microscope at NBS [40] to provide known values for each linewidth or line spacing.

A. Filar Eyepiece

The oldest of the linewidth-measurement attachments is the filar or micrometer eyepiece. Historically, it has been used with an optical microscope to measure everything from biological and metallurgical specimens to IC reticles, photomasks, and wafers. Most discussions of the filar eyepiece found in the literature deal with the details of line-scale calibration of the micrometer screw but omit any discussion of edge-detection criteria

[5] These procedures are being developed with the cooperation of Committee F-1 on Electronics of the American Society for Testing and Materials. When fully developed and approved, they will be available as ANSI/ASTM standards.

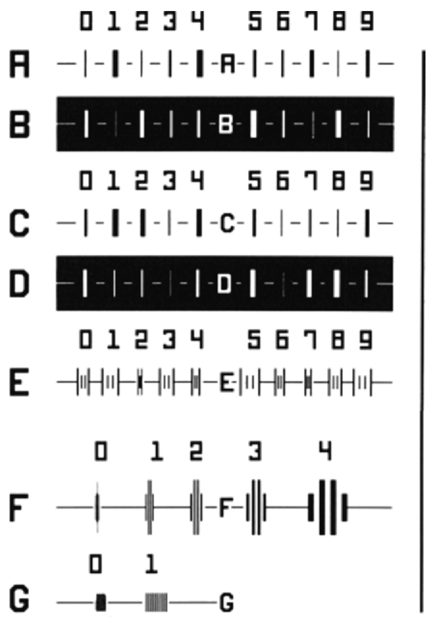


Fig. 20. Pattern for SRM 474, Optical Microscope Linewidth Measurement Standard. The scale of the illustration may be estimated by referring to pattern F4, which consists of nominally 10-μm-wide lines and spaces.

or the accuracy and precision with which width, as opposed to pitch or line spacing, can be measured.

All of the precautions associated with line-scale measurements apply to linewidth measurements as well. The eyepiece should be of the compensating type, which corrects for chromatic and other aberrations. For visual measurements, a total magnification of 1000× and an N.A. of 0.65 or greater are recommended. The measurements should always be made by moving the cross hair in one direction, starting to one side of the line, and moving in a continuous motion without dithering at the edge.

In addition, a repeatable edge-detection criterion must be used. The conventional fine black line is the most difficult to set on an edge repeatably since the cross hair blocks out the edge detail. Other types of fiducial lines available include dashed or double lines. In any case, all types depend on operator judgment of where to place the cross hair with respect to the edge. Since it is impossible to position the cross hair on a specified threshold, the method used for edge detection should be selected for a given line object and material to yield the most repeatable linewidth measurement. The system should then be calibrated for each operator to correct for linewidth errors using a set of known linewidths of the same materials as the unknown linewidths to be measured.

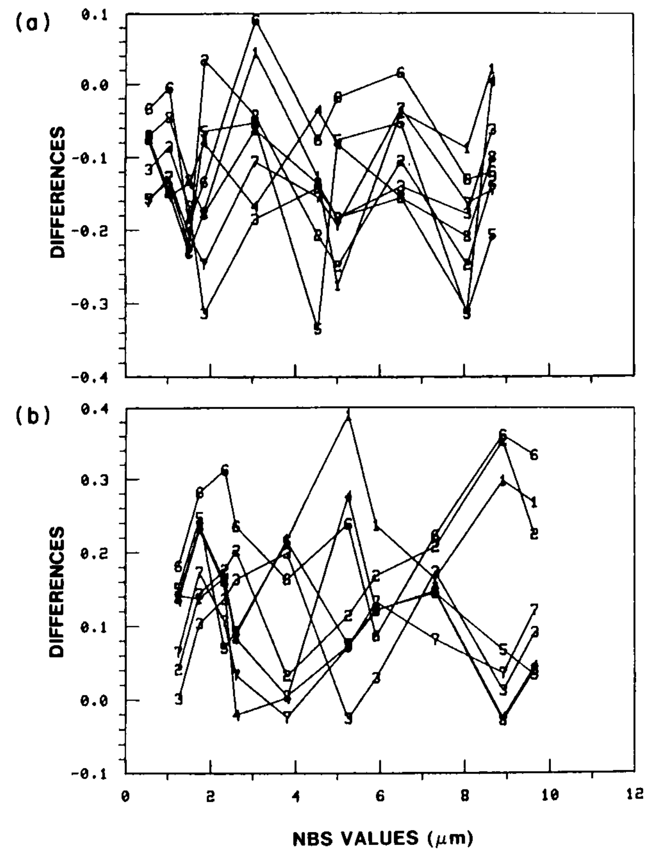


Fig. 21. Differences in linewidth measurements (measured values minus NBS values, in micrometers) for an optical filar eyepiece after calibration for line scale only. These data were obtained from an AR-chromium photomask. The numbers on the plots refer to separate data sets (a, opaque lines; b, spaces).

An example of such calibration data is shown in Fig. 21. The numbers on the curves indicate different sets of data taken at different times. Typical filar eyepieces available at the time of this study showed a precision of ± 0.30 μm or larger. Polarity-dependent errors from use of an incorrect optical threshold vary somewhat with operator and type of filar used and must be determined for each system/operator combination. For the system referred to in Fig. 21, a correction of approximately $+0.10$ μm for opaque lines and -0.15 μm for spaces should be applied. However, since the random error associated with a single measurement is larger than these corrections, the correction can only be meaningfully applied to a measurement based on an average of at least four independent measurements. If time allotted to critical dimension measurements allows for only a single measurement of an unknown linewidth, the system discussed here should not be used when an uncertainty of less than $\pm 0.3\ \mu m$ is desired.

B. Image-Shearing Eyepiece

In comparison with filar eyepieces, image-shearing measurement techniques in optical microscopy have been in use less than 25 years [3]. The superiority of an image-shearing eyepiece to a filar eyepiece has long been recognized by the IC industry.[6] The improvement in precision achieved with an image-shearing eyepiece is principally due to the use of a more repeatable edge-detection criterion.

Once the microscope is aligned and set up for Kohler illumination and adjusted for an R value of 0.67, measurements in a conventional image-shearing system should be made by first separating the two images in the counterclockwise direction and then making the linewidth measurement with a continuous motion in the clockwise direction, without dithering at the line edge. The most accurate measurements are made using the double-shear method, replacing the dichroic with a green filter such as that discussed earlier in connection with primary calibrations. Care should be taken to make sure that the direction of shear is perpendicular to the length of the line to be measured.

There is little, if any, discussion in the literature on the method of edge detection. The original instrument design assumed incoherent illumination, and in that case, the recommended method of edge setting, where no dark or bright band appears between the two abutting line images, corresponds to measurement at a 50% optical threshold, which (under the assumed incoherent illumination conditions) would also correspond to the true edge. Since incoherent illumination is not possible at high N.A.s with dry optics, the 50% threshold rarely corresponds to the true edge location. An additional complication in edge setting arises from the more complex edge structure resulting from partially coherent illumination and the optical phase difference at the line edge. Therefore, it is best to choose a repeatable edge-setting criterion suitable for the type of material. The edge-detection errors associated with the use of the wrong threshold may then be corrected by calibration using a set of known linewidths.

An example of calibration data for an image-shearing system is shown in Fig. 22. There is some variation from one system to another because of

[6] Recent improvements, including the incorporation of a strain gauge in the micrometer screw mechanism, have improved the precision of the method further. However, these improvements are not reflected in this discussion since such instruments were not available at the time of the field study on which the present discussion is based.

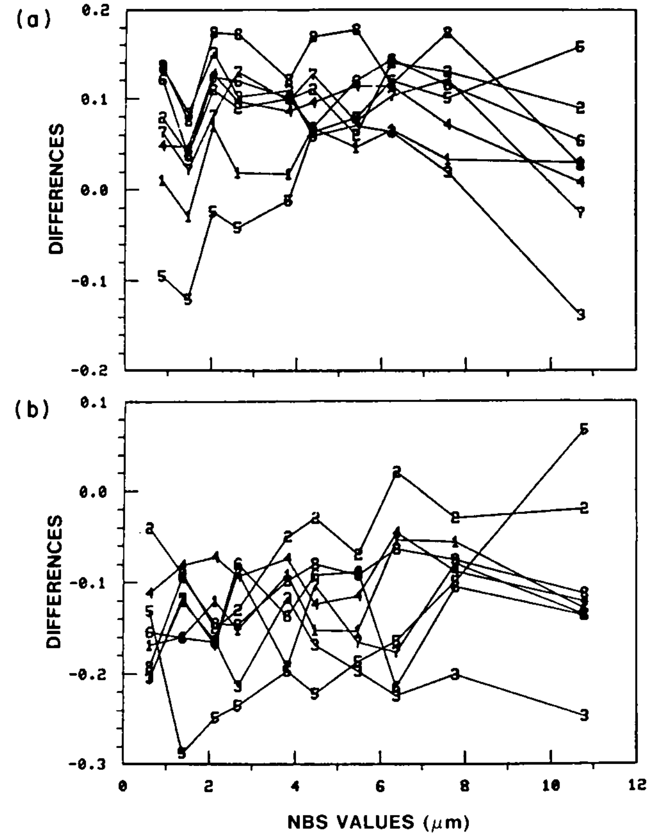


Fig. 22. Differences in linewidth measurements (measured values minus NBS values, in micrometers) for an optical image-shearing system after calibration for line scale only. These data were obtained from an AR-chromium photomask. The numbers on the plots refer to separate data sets (a, opaque lines; b, spaces).

operator-dependent edge setting and mechanical and optical variations in systems. For the system shown, the precision is approximately ± 0.15 μm, and the polarity-dependent errors require corrections of approximately -0.10 μm for opaque lines and $+0.13$ μm for spaces. With calibration, this system would yield an uncertainty in linewidth of approximately ± 0.15 μm, about half that of the filar eyepiece discussed earlier.

C. Image Scanning

Optical image-scanning linewidth-measurement systems are found less frequently in production environments than image-shearing systems. This is probably due to the slowness associated with scanning and the complexity of early instruments, which required extensive operator training and maintenance. With improvements in instrumentation and automation, this situation is changing. An image-scanning system has the advantage of being able to vary the edge-detection threshold to accommodate differing materials, thereby reducing systematic errors and, in some cases, eliminating the need for corrections to the measurements. However, it must be emphasized that control measurements are still required to maintain reproducibility.

In an image-scanning system, the visual eyepiece is replaced with a magnifying relay lens, which projects the image onto a scanning slit or vidicon. In high-resolution applications, as in the case of optical image scanning of micrometer-sized lines, the photodetector placed behind the scanning slit is usually a photomultiplier tube because of the small scanning slit dimensions (the effective slit width must be $\frac{1}{8}$ or less the Airy disk diameter of the image optics if it is not to influence the image structure) and low-light level. Either the scanning slit may be moved across the image or the object stage moved while the scanning slit remains fixed. In some systems, the sample is moved and the scanning slit dithered to provide an ac output signal [6]. A fixed stage requires better-corrected imaging optics off-axis and a flat-field corrected relay lens. However, the mechanical difficulties of moving the slit are less stringent than moving the stage. If the stage is moved, its motion must be as smooth as possible. The best mechanical ball-bearing stages are barely able to hold the focus tolerances required for high N.A. objectives. The recently developed piezo-driven flexure-pivot stage described earlier [28] is well suited for scanning small objects. With the addition of interferometric position readout and vibration isolation, ± 0.04-μm precision can be achieved with this type of stage.

In order to use the optical threshold given by Eq. (10) for edge detection, the system must have ideal edge response (see Fig. 17 for AR-chromium), the line objects must have steep edges, and the detector must be able to measure relative intensity to an accuracy of 1% or less. In addition, flare light must be less than 1% of the maximum flux used in the measurement. If these requirements cannot be met (e.g., too much flare light), the optical threshold used for measurements should be set so that it yields an accurate measurement of a known linewidth of the same material. In this case, calibration, as for filar and image-shearing systems, is necessary over the range of linewidths of interest.

D. Video Systems

Video systems consisting of a vidicon, or similar device, camera, and monitor are frequently added to linewidth-measurement systems because of the greater operator comfort associated with viewing a TV monitor for many hours as compared with looking through a microscope. It can be stated positively that the addition of a conventional video system to the microscope does not improve the quality of the measurement system. In fact, great care must be taken in setting up such a system in order to prevent degradation of the image profiles and loss of precision. The chief problems result from poor sensitivity and the inherent limitation on the white-to-black levels that can be reproduced. Electronic drift, dimensional nonlinearity, misalignment of the vidicon, and poor-quality relay optics may also degrade the accuracy and precision of the system.

The optical intensity level in the image structure corresponding to the vidicon black level is determined principally by the brightness of the lamp, the spectral bandpass of the system, the N.A. of the condenser, the sensitivity of the vidicon, and the selected video level. All of these are generally variable. However, the recommendations for making the most accurate and precise linewidth measurements discussed earlier specify the spectral bandpass and the N.A. of the condenser, leaving only the brightness of the lamp, sensitivity of the vidicon, and the video level to be adjusted. For AR-chromium photomasks, a 100–150-W tungsten–halogen lamp in conjunction with a more sensitive vidicon (1 lx or 0.1 fc minimum) allows good contrast imagery. However, day-to-day variations in linewidth measurements are certain to occur unless care is taken to set the lamp voltage and video level each day to produce the same black-and-white levels with respect to the image profile.

The best way to assure a satisfactory image profile is to view the video image profile on a cathode-ray tube (CRT). In video filar and image-shearing systems, it is the display signal that should be viewed on the CRT, whereas in a video image-scanning system, it is the video signal from the camera used to make the measurement that should be properly adjusted. For AR-chromium photomasks, T_0 should be reproduced at or above the video black level. Once the best lamp voltage and video level are established, these adjustments should be set to these same levels each day. The precision of the system will then be limited only by other variables such as electronic drift.

Typical vidicons reproduce an optical intensity range corresponding to an optical density range of approximately 1.0 D. The system cannot, therefore, reproduce all of the gray levels of a high-contrast optical image, which often exceed this optical density range. The gray level that is most important to preserve is that corresponding to the threshold T_c used for accurate edge detection. For video image-scanning systems, it is possible to adjust the electronic threshold used for measurement to yield accurate linewidth measurements provided that the optical threshold T_c lies within the linear response range.

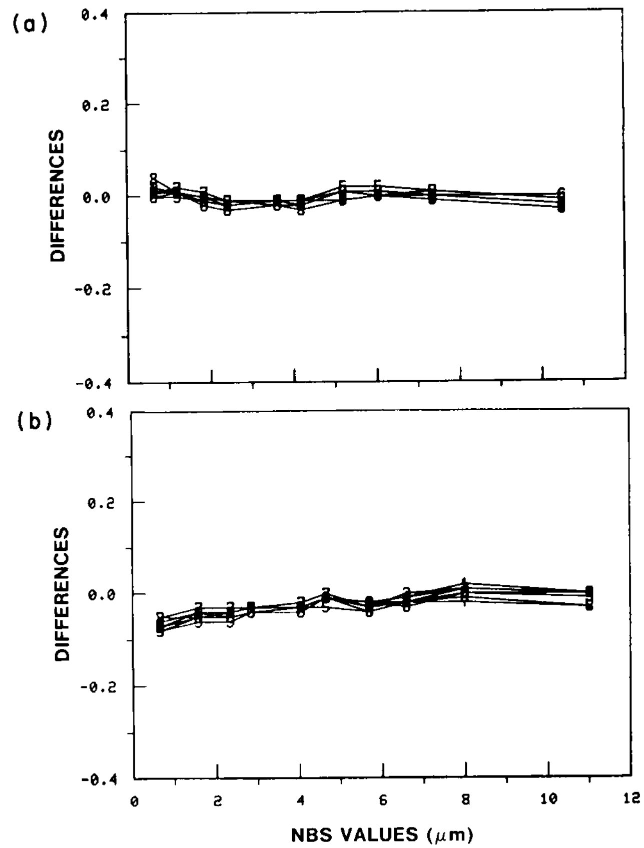


Fig. 23. Differences in linewidth measurements (measured values minus NBS values, in micrometers) for a video image-scanning system after adjustment to NBS linewidth and line-scale values. These data were obtained from an AR-chromium photomask. The numbers on the plots refer to separate data sets (a, opaque lines; b, spaces).

Figure 23 shows calibration data for a video image-scanning system with a precision of ± 0.03 μm, which has been adjusted as discussed earlier. Although this system yields very good results, it should be emphasized that extremely poor accuracy and precision can result from improper adjustment of the video level with respect to lamp voltage and other parameters as previously discussed.

VII. LINEWIDTH MEASUREMENTS ON OTHER MATERIALS

The AR-chromium photomask represents an ideal material on which to make linewidth measurements because of the high contrast, low surface reflection (which keeps scattered light in the microscope to a minimum), and excellent edge quality. There is also a necessity, however, to make measurements on other types of photomasks, such as bright chromium and semitransparent ("see-through") materials, and in patterns in resist, silicon dioxide, and metal films on silicon wafers.

A. Other Photomask Materials

Measurement of linewidths on semitransparent photomasks such as silicon and iron oxide is hampered by the dark banding that occurs along the edges. This banding arises from the combination of low contrast and optical path difference introduced by the silicon or iron oxide in conjunction with the partial coherence of the illumination. Although the edge-detection threshold of Eq. (10), based on the simple model that assumes vertical material edges, is appropriate for measurements on this type of photomask, the accuracy of the measurement is limited by the quality of the line edges and the ability to achieve reproducible and accurate focus. For systems that do not produce an optical intensity profile from which to measure linewidth using the proper threshold, the resulting measurement errors for a given high-N.A. optical system are usually independent of linewidth but vary with background transmittance (or reflectance) and phase. Since these errors are different from those associated with measurements on AR-chromium photomasks, they can only be corrected for by calibration using a known linewidth sample with the same contrast and phase as the mask being measured [16]. Further, when semitransparent materials are viewed in either reflected or transmitted illumination, thin-film effects cause an additional problem. The reflectance (or transmittance) and phase are strongly wavelength-dependent, and a spectral bandpass much narrower than that resulting from a Wratten 60 filter[7] is required to avoid degradation of the edge response.

Bright chromium photomasks in use in the IC industry differ from AR-chromium both because of their higher transmittance and, therefore, greater sensitivity to phase and because of their higher reflectivity, which tends to produce a greater amount of scattered light in the optical systems. Both effects change the threshold required for accurate edge detection. Theoretically, if the change in threshold is sufficiently small ($<5\%$), an AR-chromium mask may be used for calibration with only a slight loss in accuracy. Because the change in threshold is dependent upon the microscope as well as the photomask material, experimental

[7] See footnote 2 on p. 319.

data, not yet available, are needed to establish the accuracy of the transfer of measurements from AR-chromium to bright chromium.

B. Reflected-Light Measurements

Measurements on photomasks may be made with either transmitted or reflected light. In general, it is preferable to use transmitted light because of less stringent spectral bandpass requirements and lower sensitivity of linewidth to flare, thickness variations, and surface contamination. However, measurements on silicon wafers must, of course, be made with reflected light. It is by no means obvious at first glance that the features of partially coherent imaging (including phase, focus, and edge detection) apply equally well to transmitted-light measurements on photomasks and reflected-light measurements on wafers. Reflection microdensitometry and scanning microphotometry have been traditionally treated as separate disciplines; generally, both have been limited to either (1) wide-field, matched-aperture ($R = 1$) imaging, (2) dark-field imaging, or (3) laser scanning with a diffraction spot. Each of these has a serious limitation: (1) lack of analytic edge-detection algorithms (for matched aperture imaging), (2) extremely low-imaging light levels (for dark-field imaging), and (3) the presence of flare light, the difficulty of producing submicrometer scanning spots comparable to the Airy disk diameter of a 0.9-N.A. objective, and the effects of high angles of incidence on thin films (for laser scanning). For these reasons, it was concluded that none of these methods was suitable for accurate measurement of linewidth in reflected light. Instead, a system paralleling that used for the transmitted-light case was chosen.

The measurement system that has evolved at NBS [17,18,41] using this approach is shown schematically in Fig. 24. The system is a modified bright-field, reflected-light, image-scanning microscope employing a 1.4-W krypton laser illumination source at 530 nm. This wavelength was chosen because of the known diffraction-limited quality of the optical system at this wavelength. In contrast to commercial laser spot-scanning systems, this system illuminates a much larger specimen area, 30–50-μm radius, which, in turn, is imaged onto a stationary scanning slit. A rotating ground-glass disk is used to reduce the excessive spatial coherence of the laser beam, which would produce undesirable speckle and interference effects. The illuminating system is then used to regenerate the desired degree of spatial coherence determined by the coherence parameter R, which should be less than or equal to 0.25. The effective scanning slit width projected onto the sample is 0.2 μm and the N.A. of the objective lens is 0.85. As with the transmitted-light system, the sample is moved

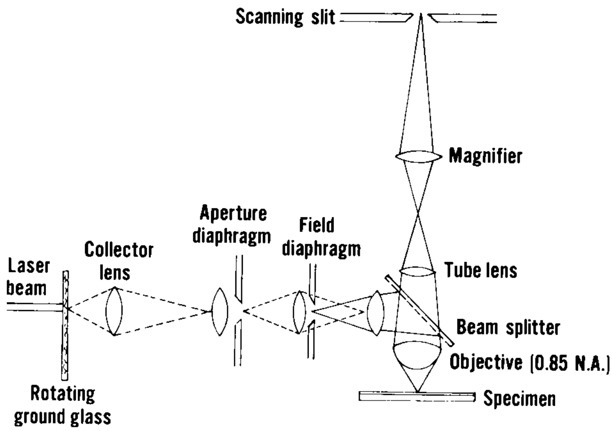


Fig. 24. Ray path for reflected-light laser-scanning microscope system.

via a flexure-pivot stage [28] with stage displacement in one dimension determined by a laser interferometer; a piezo-focusing mechanism allows focus positioning to better than 0.05 μm, and the entire system is mounted on a vibration-isolation table in a temperature-controlled environment (22° ± 1° C).

The principal differences between the NBS system and conventional bright-field, reflected-light microscope systems are: (1) the monochromatic illumination, (2) the choice of coherence parameter equal to one-fourth or less, (3) the accuracy of optical alignment and focus, (4) the vibration-isolation system, and (5) the photometric accuracy (better than 1%). In addition, the NBS system is sufficiently versatile as a research tool to allow variation of light sources and other optical system parameters.

The edge-detection threshold of Eq. (10), based on the simple model that assumes vertical material edges, is also appropriate for measurements of lines etched in thin (<200 nm) silicon dioxide films on silicon wafers. For this case, T_0 and ϕ_0 are uniquely defined only for a single wavelength (monochromatic illumination) because of the strong dependence of both the reflectance and phase on wavelength. For the silicon dioxide layer, the curves for the reflectance and phase change on reflection are periodic with a period of approximately 180 nm. When normalized relative to the recessed silicon base, the periodicity of the reflectance curves remains unchanged, although the phase curve changes dramatically, as illustrated in Fig. 25 for the case of normal incidence and a wavelength of 530 nm. These reflectance and phase curves are calculated from the Fresnel equations [26, p. 632] and normalized assuming that the silicon dioxide layer is etched through to the silicon substrate. Although,

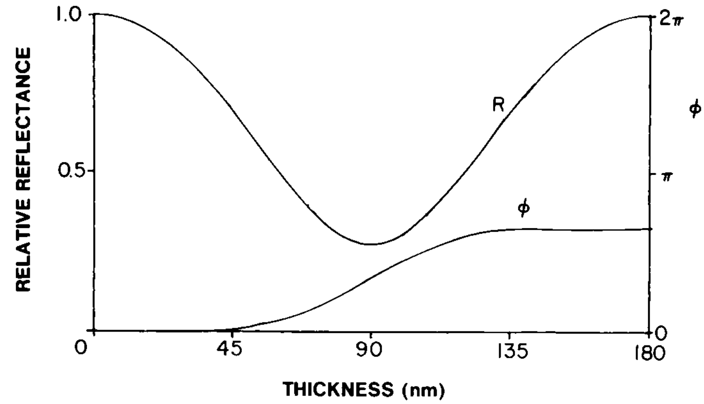


Fig. 25. Relative reflectance I_0/I_M (curve R) and phase difference ϕ (curve ϕ) for silicon dioxide on silicon calculated from the Fresnel equations for varying thicknesses of silicon dioxide for monochromatic illumination (530 nm) at normal incidence. Curves are normalized with respect to the silicon substrate.

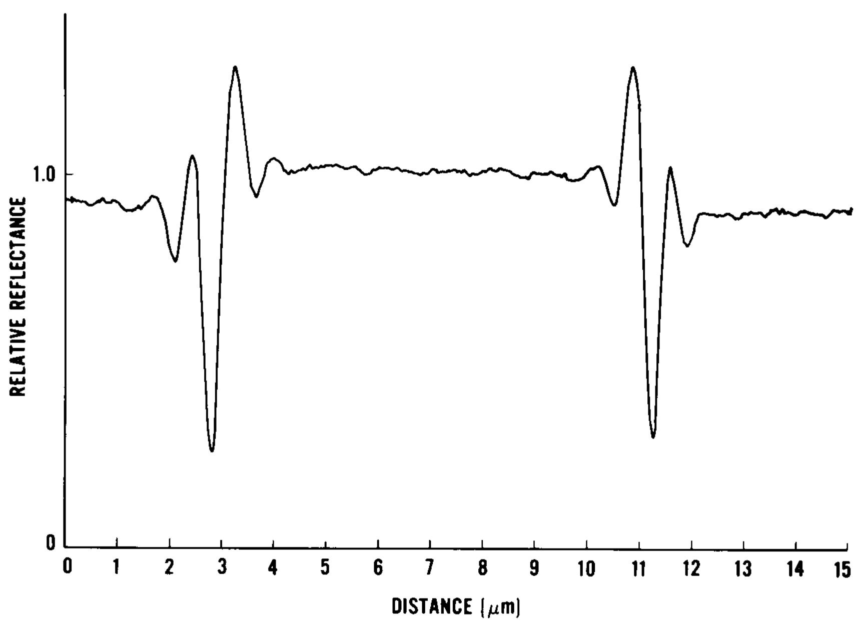


Fig. 26. Experimental image profile from reflected-light laser-scanning system for monochromatic illumination (530 nm) for silicon dioxide (~180-nm-thick layer) on silicon.

in practice, a thin silicon dioxide layer forms in the etched region, inclusion of this layer in the calculations would not qualitatively change the curves.

Figure 26 illustrates the image associated with lines etched in a 180-nm-thick layer of silicon dioxide on silicon. Figure 27 shows a comparison of the theoretical and experimentally measured image profiles of a nominally 4-μm-wide window etched in a thin silicon dioxide layer on silicon. The line was made by contact printing one of the AR-chromium photoplates discussed in Section VI onto a 150-nm-thick layer of silicon dioxide thermally grown on a silicon wafer. The oxide was plasma etched to obtain the steepest possible material edges. The relative phase is determined from the relative reflectance at the minimum of the measured image profile $I(x_0)$. It is necessary to establish ϕ_0 only to an accuracy of $\pi/10$, so graphical determination using Fig. 28 [17] is adequate. The linewidth is then determined from the measured profile using the threshold given by Eq. (10). The theoretical curve is calculated from these parameters and those of the optical system by Kintner's method [27]. Based upon the excellent agreement obtained for this and other samples, the system appears to be capable of measurements of linewidths as small as 0.5 μm with an accuracy equivalent to or better than that obtained in measurements on AR-chromium photomasks.

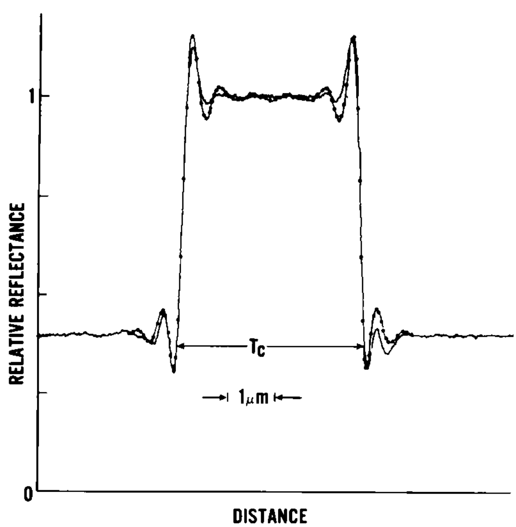


Fig. 27. Comparison of experimental (—) and theoretical (●) image profiles for a window etched in a 150-nm-thick layer of silicon dioxide on silicon (0.85 objective N.A., 0.2 condenser N.A., and 530-nm wavelength).

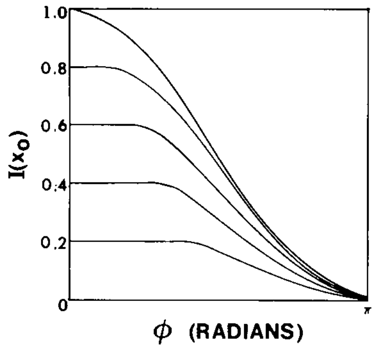


Fig. 28. Relationship of the phase difference ϕ to the relative intensity at the minimum $I(x_0)$, where x_0 is the distance of the minimum from the true edge location. The curve-to-curve parameter is T_0, indicated by the asymptotic value of the curve at $\phi = 0$.

Measured image profiles are significantly affected by differences in spectral bandwidth of the illumination, coherence parameter, and scanning aperture size. Figure 29 illustrates the difference in image profiles obtained with spectrally broadband illumination using a Wratten 60 spectral filter as compared with laser illumination (530 nm). The difference is significant enough to cause errors of 0.2 μm or more if one attempts to use

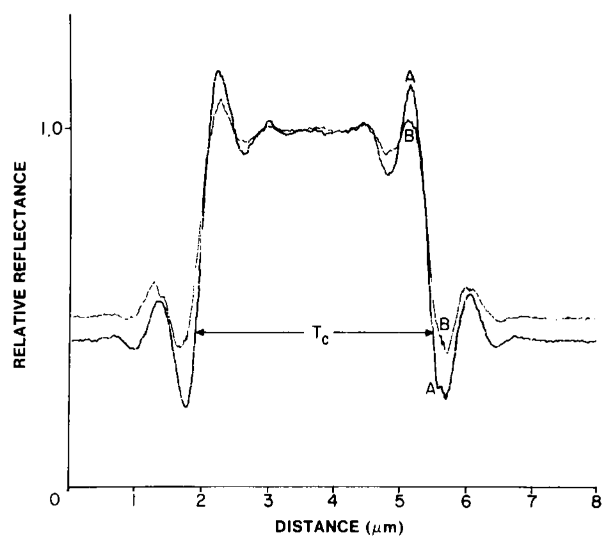


Fig. 29. Experimental image profiles comparing krypton laser illumination (530 nm) (curve A) with broadband green spectral illumination (WR 60) (curve B) (0.85 objective N.A., 0.2 condenser N.A.). The minima correspond to the dark bands seen at the line edges.

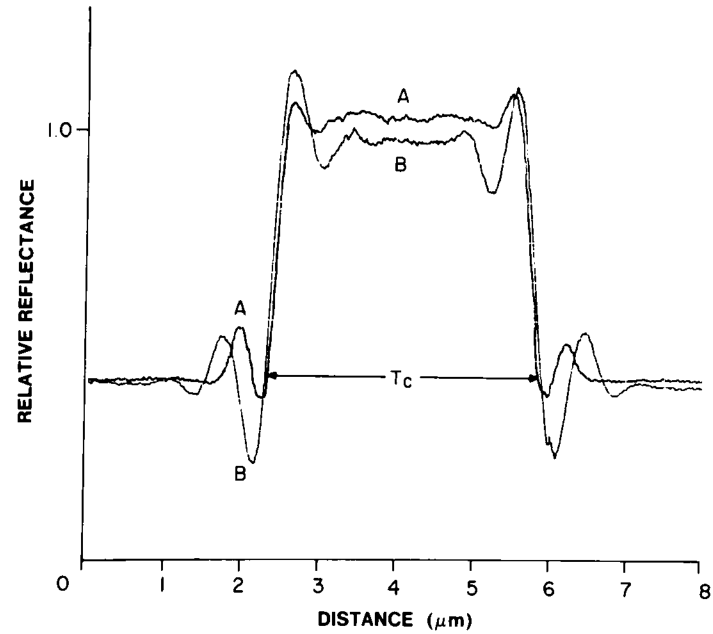


Fig. 30. Experimental image profiles for krypton laser illumination (530 nm) with coherence parameter, $R = 0.8$ (curve A) and $R = 0.2$ (curve B) (0.85 objective N.A.).

the edge-detection threshold of Eq. (10) with broadband illumination. The implication of this observed difference is that because most commercially available linewidth-measurement systems employ broadband spectral illumination, they cannot be used as primary measurement systems and, therefore, require linewidth calibration in addition to line-spacing or pitch calibration in order to correct for edge-detection errors. The difference between unfiltered illumination and broadband green is less significant, although it may vary more from system to system depending upon the degree of chromatic correction in the optical system.

As expected, increasing the coherence parameter from 0.2 to 0.8 changes the profile significantly, as shown in Fig. 30; this figure also illustrates the dependence of the measured reflectance of silicon relative to silicon dioxide on the optical system geometry, an effect well known in the field of microdensitometry [42].

C. Wafer Measurements with Conventional Optical Systems

The lines on the patterned wafer just described were calibrated using the NBS laser, reflected-light, optical image-scanning linewidth-

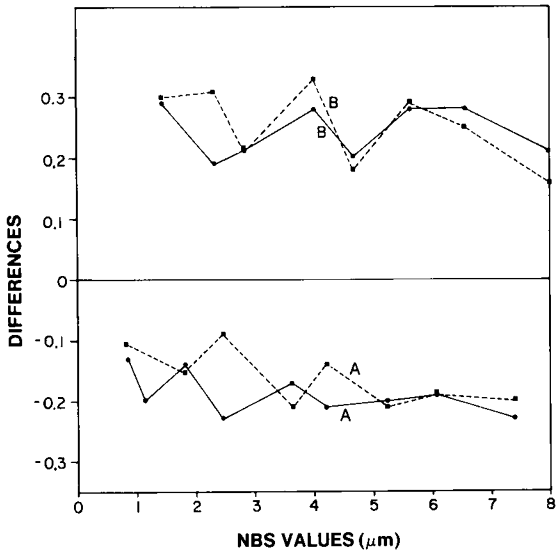


Fig. 31. Differences (measured values minus NBS values) for filar (---) and image-shearing (—) linewidth-measurement systems. Both systems used center of dark band at the line edge for measurement. These data apply only to a 150-nm-thick etched silicon dioxide layer (curves A, lines; curves B, spaces).

measurement system. This wafer was then measured on a variety of commercially available linewidth-measurement systems [41]. Some typical results are shown in Figs. 31–33. The measurements illustrated in these figures were made under more controlled conditions than is customary for measurements currently being made in the IC industry. In each case, the optical microscope was set up following the NBS-recommended procedures for photomasks, which include the use of Kohler illumination, a coherence parameter $R = \frac{2}{3}$, and broadband green (WR 60[8]) spectral illumination. Although an R value of $\frac{1}{4}$ is the best choice for wafer measurements, some older linewidth-measurement systems, which do not have mercury arc sources, may not have sufficient lamp brightness and detector sensitivity to operate at this R value. For such systems, an R value of $\frac{2}{3}$ appears to be an acceptable compromise.

For the measurements shown in Fig. 31, a filar eyepiece (dashed line) and a coincidence-setting image-shearing system with video display (solid line) were used. The video system required a mercury arc source to produce a good video image with the broadband green filter. For both of these systems, the visual center of the dark interference band along the

[8] See footnote 2 on p. 319.

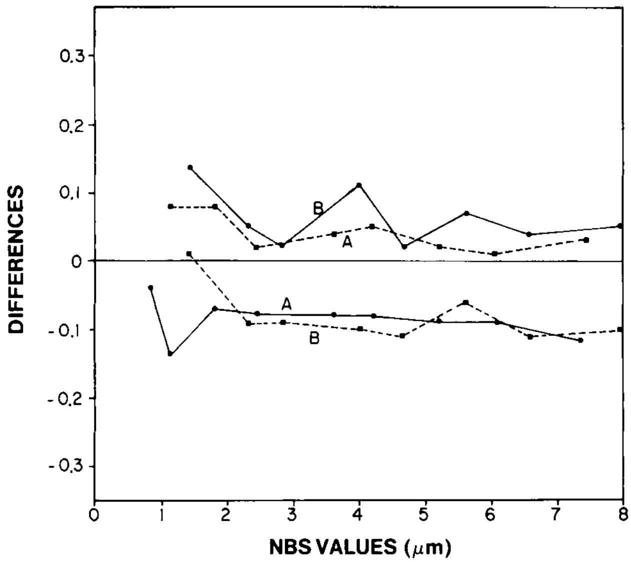


Fig. 32. Differences (measured values minus NBS values) for optical image-scanning (—) and laser scanning microspot (---) linewidth-measurement systems. These data apply only to a 150-nm-thick etched silicon dioxide layer (curves A, lines; curves B, spaces).

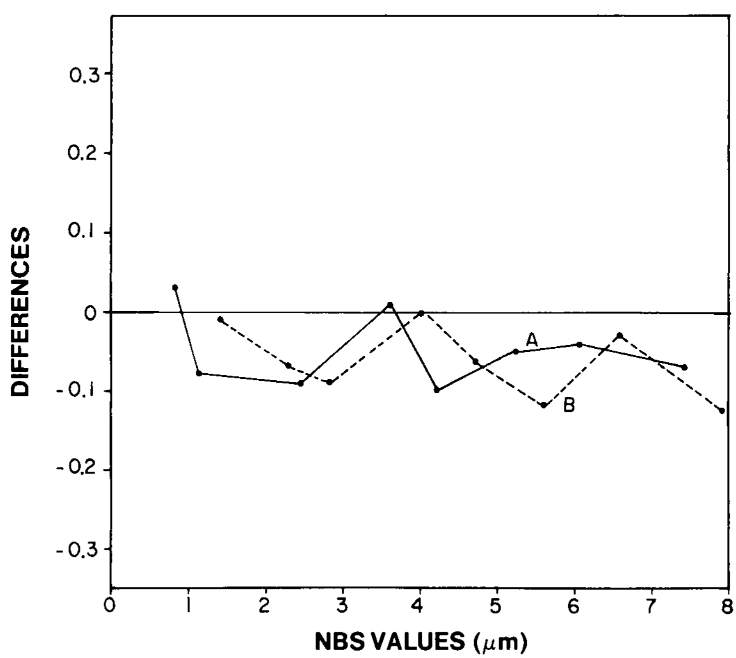


Fig. 33. Differences (measured values minus NBS values) for a video image-scanning linewidth-measurement system. These data apply only to a 150-nm-thick etched silicon dioxide layer (curve A, lines; curve B, spaces).

line edge (see Fig. 27) was used to determine edge location. Although this criterion does not yield the most accurate linewidth measurements, for both of these systems it was felt to yield the most repeatable measurements, and the data can therefore be used to determine the system calibration curves. Each of the measured linewidths shown in the figure is an average of four measurements. The precision of all the measurements is ± 0.24 μm for the optical filar and ± 0.06 μm for the image-shearing system. The latter value is smaller than was obtained for image-shearing data on photomasks (see Section VI.B); the improvement is attributed to the incorporation of a strain gauge in the shear mechanism. The principal effect illustrated is the polarity-dependet edge-detection error expected from use of the center of the dark interference band. In both cases, the data shown could be used as calibration data to correct for this known systematic error.

Data for two types of optical image-scanning systems using different edge-detection methods are shown in Fig. 32. One system (solid line) used broadband green illumination and scanned the image with a moving slit. Since only a few choices of edge-detection criteria were available for this system, the one used for photomasks appeared to be the best choice. It selected the edge at a threshold corresponding to 33% above the minimum measured reflectance. For this particular wafer, this particular edge-detection criterion is close to the true-line edge as shown in the figure. The other system (dashed line) employed laser illumination (λ = 632.8 nm) focused to a small spot on the wafer. In this system, two off-axis detectors sense edges and provide maximum signals at the line edges. This system might be described as an inverted dark-field microscope. The correspondence between an optical threshold and the edge-detection criterion used by this system is unknown. These data, however, give an idea of the relationship of this method to that of other photometric linewidth-measurement systems. The precision is estimated to be ± 0.09 μm for the image-scanning system and ± 0.02 μm for the microspot scanner. The small precision values allow the calibration curves represented by these data to be meaningfully applied as a correction to single measurements of unknown linewidths.

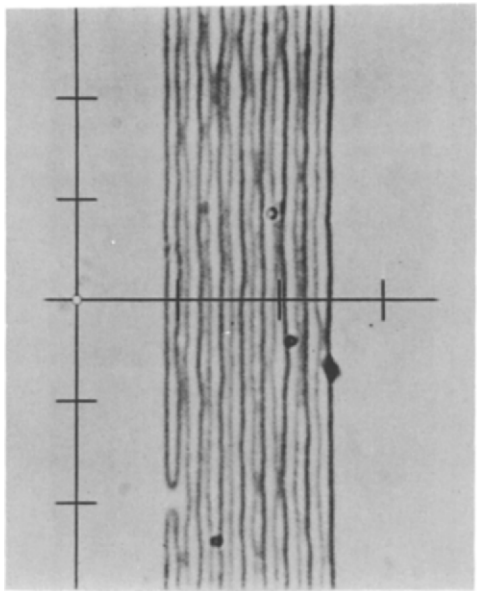
Measurements were also made on a video image-scanning system; the results are shown in Fig. 33. This system differs from all of the systems discussed earlier in that it has a continuously variable electronic threshold as well as line-to-space ratio correction circuitry. In principle, if a correspondence can be made between the electronic and optical threshold using a known linewidth, the system should be able to measure linewidth accurately. However, the image profiles produced by this system were significantly degraded because of the marginal sensitivity of the vidicon, the low reflectance, and inadequate illumination levels, which may account for the observed edge-detection errors. Since completion of the study upon which these results are based [41], video linewidth-measurement systems with significantly improved imaging have become commercially available.

D. Calibration Procedures for Wafer Measurements

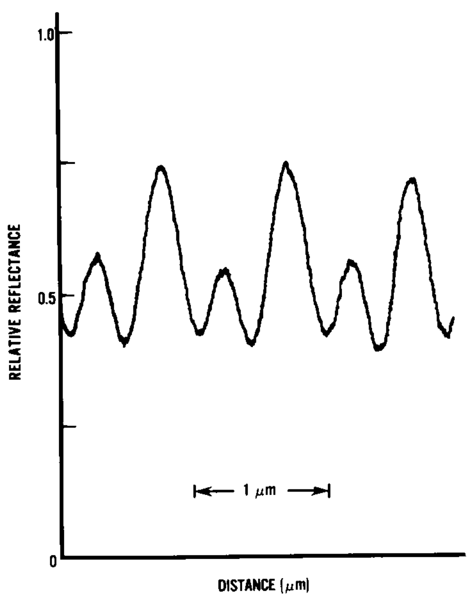
The results shown in Figs. 31–33 represent only one combination of relative reflectance and phase. In principle, since the calibration is material-dependent, a separate calibration curve is required for each material and each thickness corresponding to each step in the wafer-fabrication process where linewidth measurements are made. However, it is not feasible to provide calibration wafers for, or calibrate, linewidth-measurement systems for all combinations of relative reflectance and phase for all materials of interest. An alternative approach is to supply a small number of calibration wafers, say, five or more combinations of relative reflectance and phase, most of which correspond to different thicknesses of silicon dioxide on silicon. For making measurements of an unknown linewidth, the system would be calibrated using the calibration wafer that most closely matches the unknown in relative reflectance and phase. The maximum possible error could be determined from the system calibration data and a correction could be applied if the relative reflectance and phase of the unknown were measured. For instance, when the center of the dark band at the line edge is used for measurement, as in the case of filar and image-shearing systems, the error resulting from a mismatch in relative reflectance and/or phase can be computed [16]. The accuracy of this proposed calibration method is dependent upon the predictability and precision of the calibration curves for a given instrument and remains to be determined for systems in use in the IC industry.

E. Extensions to Patterns in Thicker Layers

It is of interest to note that image profiles of lines in transparent materials such as silicon dioxide or resist are visible well below the conventional resolution of the optical system because of the phase and coherence in the system. Figure 34 shows a photometric scan and corresponding photograph of nominal 0.5-μm lines and spaces in resist as viewed in an effectively coherent reflected-light system, illustrating the variation in linewidth visible in the line structure well below the conventional 0.5-μm resolution limit of the system [42]. Using the proper threshold and focus, accurate measurements of these lines can be made.



(a)



(b)

Fig. 34. Photomicrograph (a) and corresponding image profile (b) of nominal 0.5-μm lines and spaces in resist. Dark bands (minima) correspond to approximate location of edges.

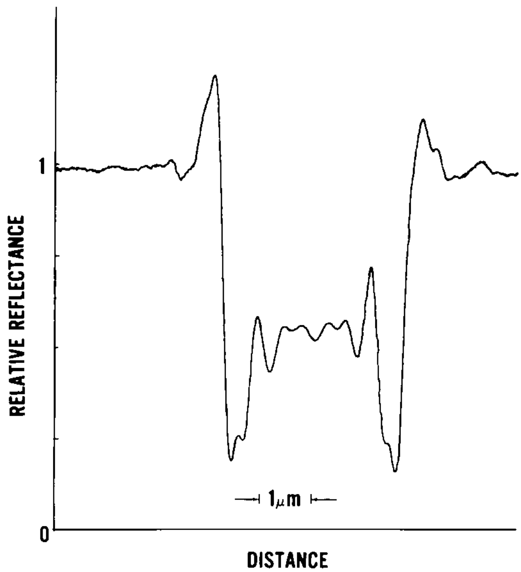


Fig. 35. Experimentally generated line-image profile for 600-nm-thick layer of etched silicon dioxide on silicon from reflected-light laser-scanning system. Compare with Fig. 27.

The limitations of scalar theory become apparent as the sample thickness is increased. Figure 35 is an experimentally measured image profile for a 600-nm-thick layer of etched silicon dioxide on silicon. Similarly, noticeable changes in the image structure due to the thickness of the etched layer are observable with layer thicknesses as small as 400 nm in transmitted light or 200 nm in reflected light. The imaging of thick layers is further complicated by the effect of nonvertical material edges and requires a more complex model of the line object than that given here. To extend the theory to thick layers, a model is being developed based on the use of waveguide analysis [43–46] to propagate an incident plane wave through the thick layer and the application of scalar theory to treat the imaging of the resulting optical intensity distribution [47].

REFERENCES

1. L. C. Martin, "The Theory of the Microscope," Chapters V and VIII. Blackie, London, 1966.
2. J. M. Jerke, Semiconductor Measurement Technology: Accurate Linewidth Measurements on Integrated-Circuit Photomasks, National Bureau of Standards Special Publication 400–43, p. 7 (February 1980).
3. J. Dyson, *J. Opt. Soc. Am.* **50** (8), 754 (1960).

4. D. A. Swyt and F. W. Rosberry, *Solid State Technol.* **20** (8), 70 (1977).
5. G. W. Deberko, unpublished paper presented at SEMICON/WEST Show, San Mateo, California (May 26, 1976).
6. P. E. Ciddor, *Metrologia* **7** (4), 135 (1971).
7. H. L. Kasdan and N. George, *Proc. Soc. Photo-Opt. Instrum. Eng. Develop. Semicond. Microlithogr.* **80,** 54 (1976).
8. A. L. Flamholz and R. S. Charsky, *Proc. Soc. Photo-Opt. Instrum. Eng. Develop. Semicond. Microlithogr. III* **135,** 120 (1978).
9. R. E. Swing, *Proc. Soc. Photo-Opt. Instrum. Eng. Develop. Semicond. Microlithogr.* **80,** 65 (1976).
10. H. R. Rottmann, *IBM J. Res. Dev.* **18** (3), 225 (1974).
11. H. R. Rottmann, *J. Appl. Photo. Eng.* **2** (3), 119 (1976).
12. D. Nyyssonen, *in* Semiconductor Measurement Technology, Progress Report (W. M. Bullis, ed.), National Bureau of Standards Spec. Publ. 400-38, p. 33 (November 1979).
13. S. Jensen and D. Swyt, *Scanning Electron Microsc.* **I,** 393 (1980).
14. J. M. Jerke, A. W. Hartman, D. Nyyssonen, R. E. Swing, R. D. Young, and W. J. Keery, *Proc. Soc. Photo-Opt. Instrum. Eng. Develop. Semicond. Microlithogr. II* **100,** 37 (1977).
15. D. Nyyssonen, *Appl. Opt.* **16,** 2223 (1977).
16. D. Nyyssonen, *Proc. Soc. Photo-Opt. Instrum. Eng. Develop. Semicond. Microlithogr. II* **100,** 127 (1977).
17. D. Nyyssonen, *Proc. Soc. Photo-Opt. Instrum. Eng. Develop. Semicond. Microlithogr. III* **135,** 115 (1978).
18. D. Nyyssonen, *Proc. Soc. Photo-Opt. Instrum. Eng. Appl. Opt. Coherence* **194,** 34 (1979).
19. B. Richards and E. Wolf, *Proc. R. Soc. London Ser. A* **A253,** 358 (1959).
20. B. M. Watrasiewicz, *Opt. Acta* **12,** 167 (1965).
21. B. M. Watrasiewicz, *Proc. R. Soc. London Ser. A* **293,** 391 (1966).
22. W. N. Charman, *J. Opt. Soc. Am.* **53,** 410 (1963).
23. W. N. Charman, *J. Opt. Soc. Am.* **53,** 415 (1963).
24. R. E. Kinzly, *J. Opt. Soc. Am.* **55,** 1002 (1965).
25. J. W. Goodman, "Introduction to Fourier Optics," Chapter 5. McGraw-Hill, New York, 1968.
26. M. Born and E. Wolf, "Principles of Optics," 5th ed. Pergamon, Oxford, 1975.
27. E. C. Kintner, *Appl. Opt.* **17,** 2747 (1978).
28. F. W. Scire and E. C. Teague, *Rev. Sci. Instrum.* **49,** 1735 (1978).
29. R. E. Swing, *Opt. Eng.* **15,** 559 (1976).
30. H. D. Cook and L. A. Marzetta, *J. Res. Nat. Bur. Std. U.S.A.* **65C,** 129 (1961).
31. SRM 474, Optical Microscope Linewidth-Measurement Standard, available from the National Bureau of Standards Office of Standard Reference Materials, Washington, D. C. 20234.
32. S. Jensen, G. Hembree, J. Marchiando, and D. Swyt, *Proc. Soc. Photo-Opt. Instrum. Eng. Develop. Semicond. Microlithogr. VI* **275,** 100 (1981).
33. A. Kuni, Y. Kembo, K. Yoshimura, and N. Aoki, *Proc. Microelectron. Measurement Technol. Seminar,* p. 67. Benwill, Boston, Massachusetts, 1980.
34. K. Mizukami, Y. Hisamoto, Y. Wade, and M. Migataka, *Proc. Microelectron. Measurement Technol. Seminar,* p. 97. Benwill, Boston, Massachusetts, 1980.
35. K. Nakazawa, S. Yoshida, A. Tanemoto, S. Murakami, and T. Yamaguchi, *Proc. Microelectron. Seminar—Interface '80,* Publication No. E,G-130. Eastman Kodak Co., Rochester, New York, 1981.

36. G. Hembree, S. Jensen, and J. Marchiando, *Proc. Ann. Meeting Electron Microsc. Soc. Am., 38th* (G. Bailey, ed.), p. 312. Claitor's Publ., Baton Rouge, Louisiana, 1980.
37. I. Adesida and T. E. Everhart, "Extended Abstracts 80-1," p. 643. The Electrochemical Society, Pennington, New Jersey, 1980.
38. D. A. Swyt, F. W. Rosberry, and D. Nyyssonen, *Proc. Microelectron. Seminar—Interface '77*, p. 131. Eastman Kodak, Rochester, New York, 1977.
39. J. M. Jerke, M. C. Croarkin, and R. N. Varner, Semiconductor Measurement Technology: Interlaboratory Study for Antireflective Chromium Photomasks. National Bureau of Standards Spec. Publ. (in preparation).
40. D. A. Swyt, *Proc. Soc. Photo-Opt. Instrum. Eng. Effective Utilizat. Opt. Quality Assurance* **129,** 98 (1977).
41. D. Nyyssonen, *Proc. Soc. Photo-Opt. Instrum. Eng. Develop. Semicond. Microlithogr. V* **221,** 119 (1980).
42. T. H. James (ed.), "The Theory of the Photographic Process," 3rd ed., p. 483. MacMillan, New York, 1977.
43. F. G. Kaspar, *J. Opt. Soc. Am.* **63,** 37 (1973).
44. F. G. Kaspar, *J. Opt. Soc. Am.* **64,** 1623 (1974).
45. C. B. Burckhardt, *J. Opt. Soc. Am.* **56,** 1502 (1966).
46. D. Nyyssonen, High Resolution Microdensitometry of Photographic Emulsions. Ph.D. Thesis, Univ. of Rochester, Rochester, New York (1975).
47. D. Nyyssonen, Theory of Optical Edge Detection (to be published).